

Data Communications and Networking Fourth Edition

Forouzan

# Chapter 2

# Network Models



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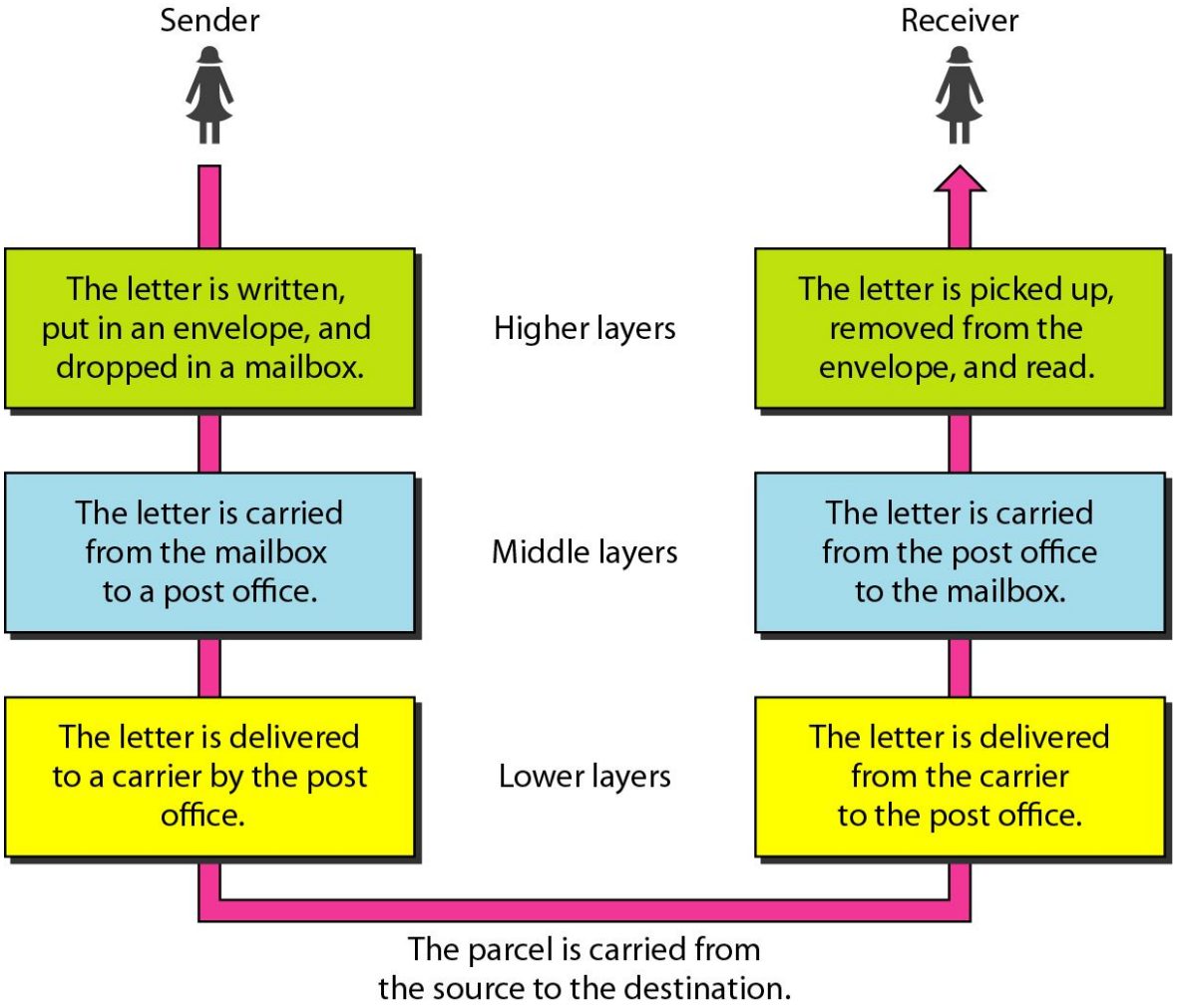
# 2-1 LAYERED TASKS

*We use the concept of layers in our daily life. As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.*

***Topics discussed in this section:***

**Sender, Receiver, and Carrier**
**Hierarchy**

## Figure 2.1 Tasks involved in sending a letter

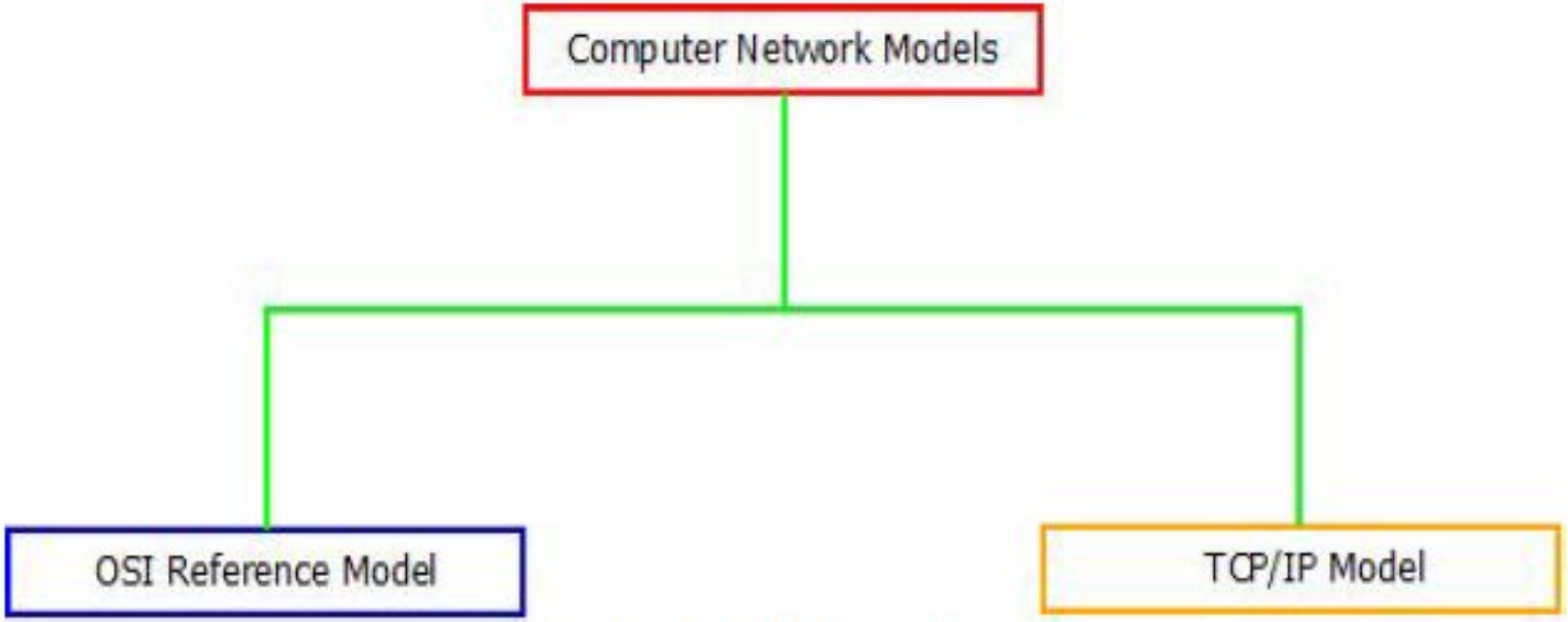


# Computer Network Models

- For data communication to take place and two or more users can transmit data from one to other, a systematic approach is required. This approach enables users to communicate and transmit data through efficient and ordered path. It is implemented using models in computer networks and are known as computer network models.

- Computer network models are responsible for establishing a connection among the sender and receiver and transmitting the data in a smooth manner respectively.

- There are two computer network models i.e. OSI Model and TCP/IP Model on which the whole data communication process relies.




**Computer Network Models**

# 2-2 THE OSI MODEL

**Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s.**

*Note*

**ISO is the organization.**
**OSI is the model.**

# The OSI Reference Model

- **The OSI Model is one of the general purpose networking or communication model among computer network models, which is responsible for establishing connection in an open manner between all the communicable devices present across the globe.**

- **OSI stands for "Open System Interconnection" and the name of this reference model was given by an organization known as "International Organization for Standardization". The ISO is responsible for generating and promoting industrial and commercial standards applicable for all the users or universally.**

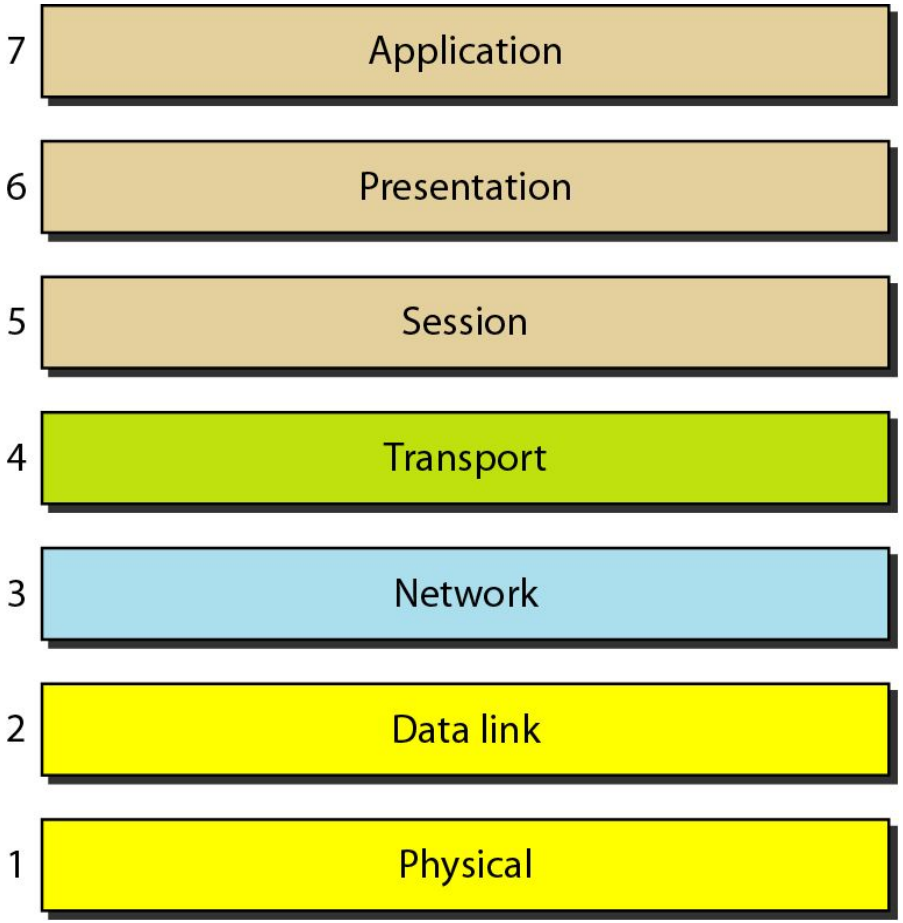
- **OSI model is called as "Open Source" because of its "fit anywhere" ability. Any connection can be established using the OSI model unless and until any protocols are not used as OSI model does not support protocol establishment. It runs without the use of protocols. All the networking devices which are open for communication can be connected together for data communication through the OSI model.**

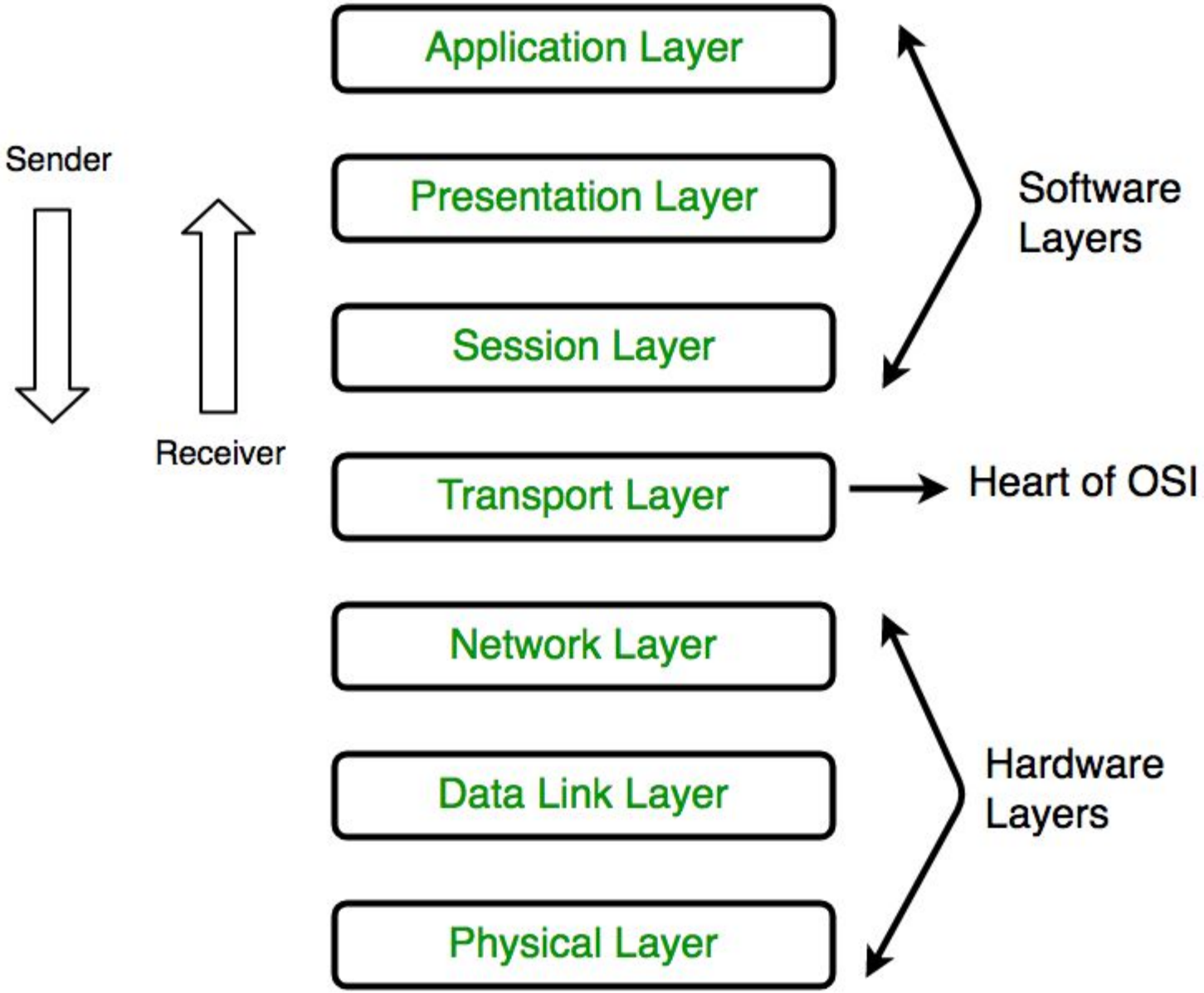
- **Apart from OSI Model, another computer network models which is widely used is TCP/IP Model.**

- **OSI model having a layered architecture, allows easy data communication as each layer has predefined structured and functionalities.**

- **The functionalities are different for each layer and thus when combined together forms the OSI Model. There are in total seven layers in general purpose OSI model.**

## Figure 2.2 *Seven layers of the OSI model*




Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer
Physical Layer
Sender
Receiver
Software Layers
Heart of OSI
Hardware Layers

# OSI Model



Open System Interconnection Model

or

OSI seven layer model

or

Open System

Enable functionality and interoperability between different vendors and connectors

# OSI Model

| | |
|---|---|
| 7 | Application |
| 6 | Presentation |
| 5 | Session |
| 4 | Transport |
| 3 | Network |
| 2 | Data Link |
| 1 | Physical |

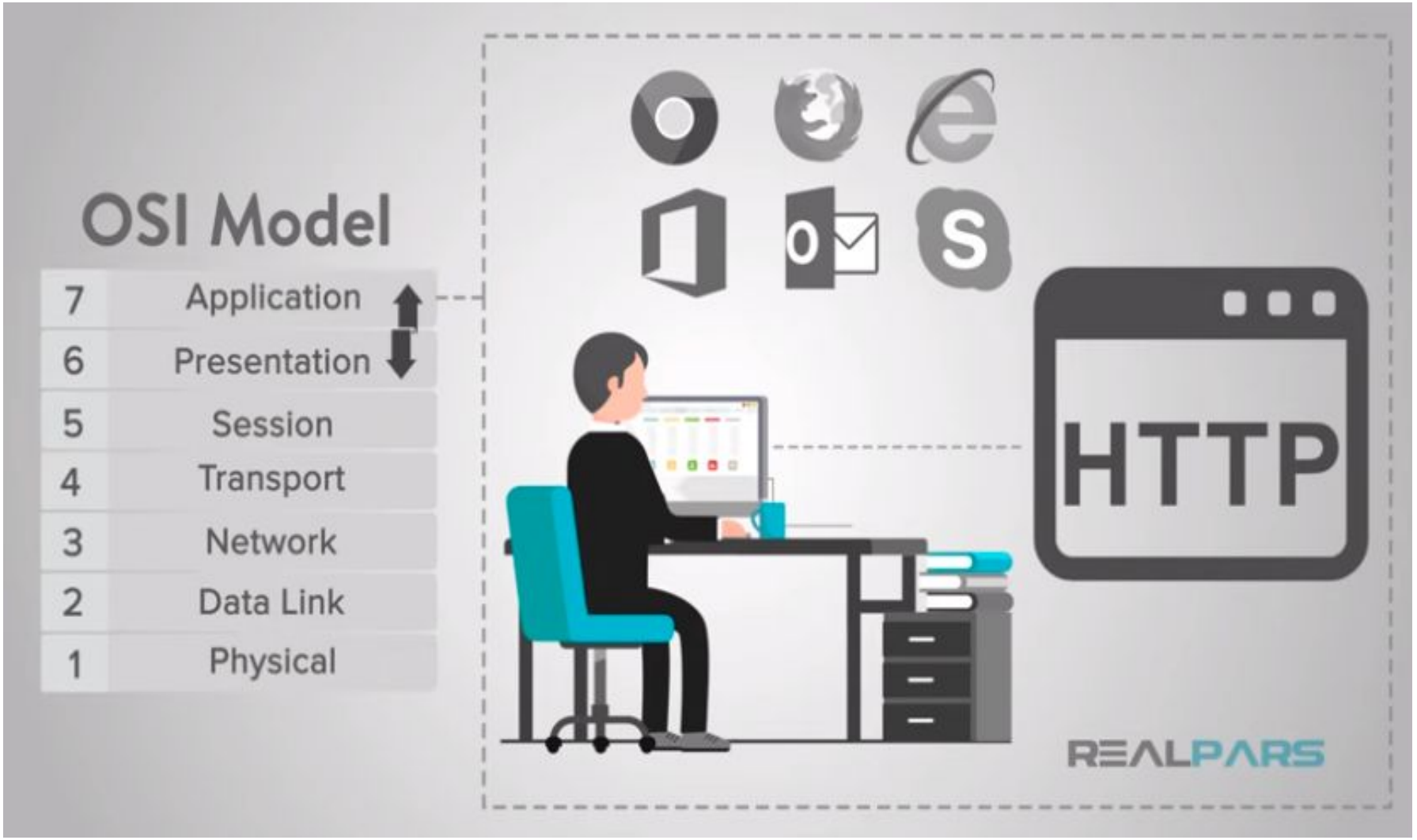
- Purpose of Networking:
  - **Allow two hosts to share data with one another**

- Hosts must follow a set of rules
  - Example: English language has rules
    Spanish language has rules

- The rules for Networking are divided into seven layers:
  - OSI Model



- Human Body is made up various systems:
  - Skeletal, Respiratory, Nervous, Cardiovascular, Muscular, etc
  - If all these systems are functioning, the human lives


OSI Model
7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical
HTTP
REALPARS

# Application Layer: Network Applications










File Transfer ✓ Web Surfing ✓ Emails ✓ Virtual Terminals ✓





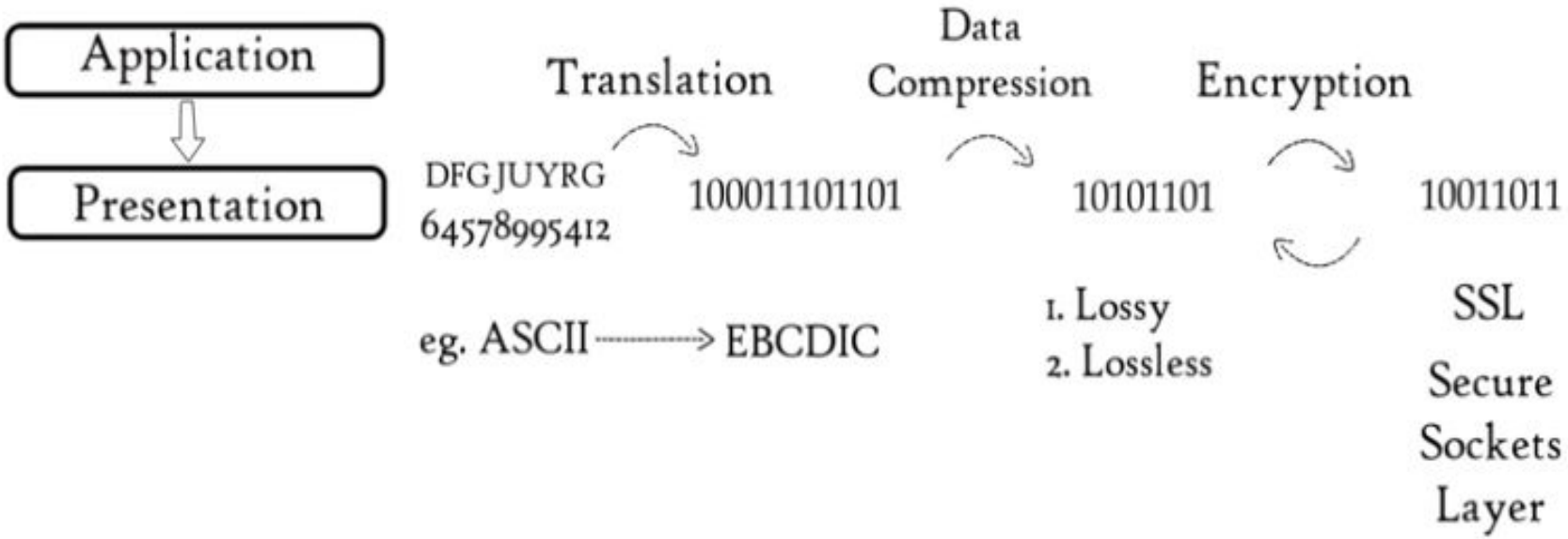


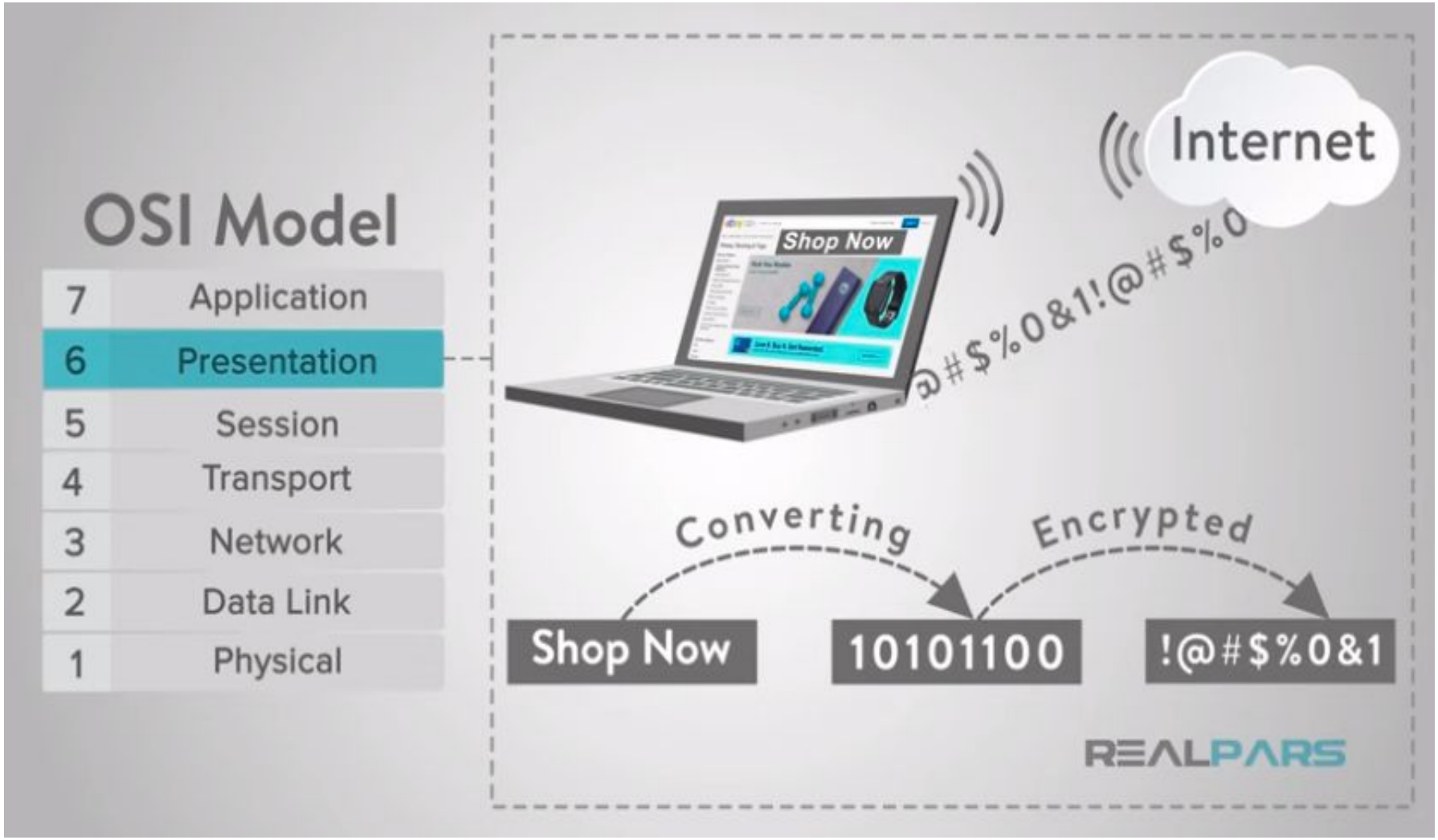


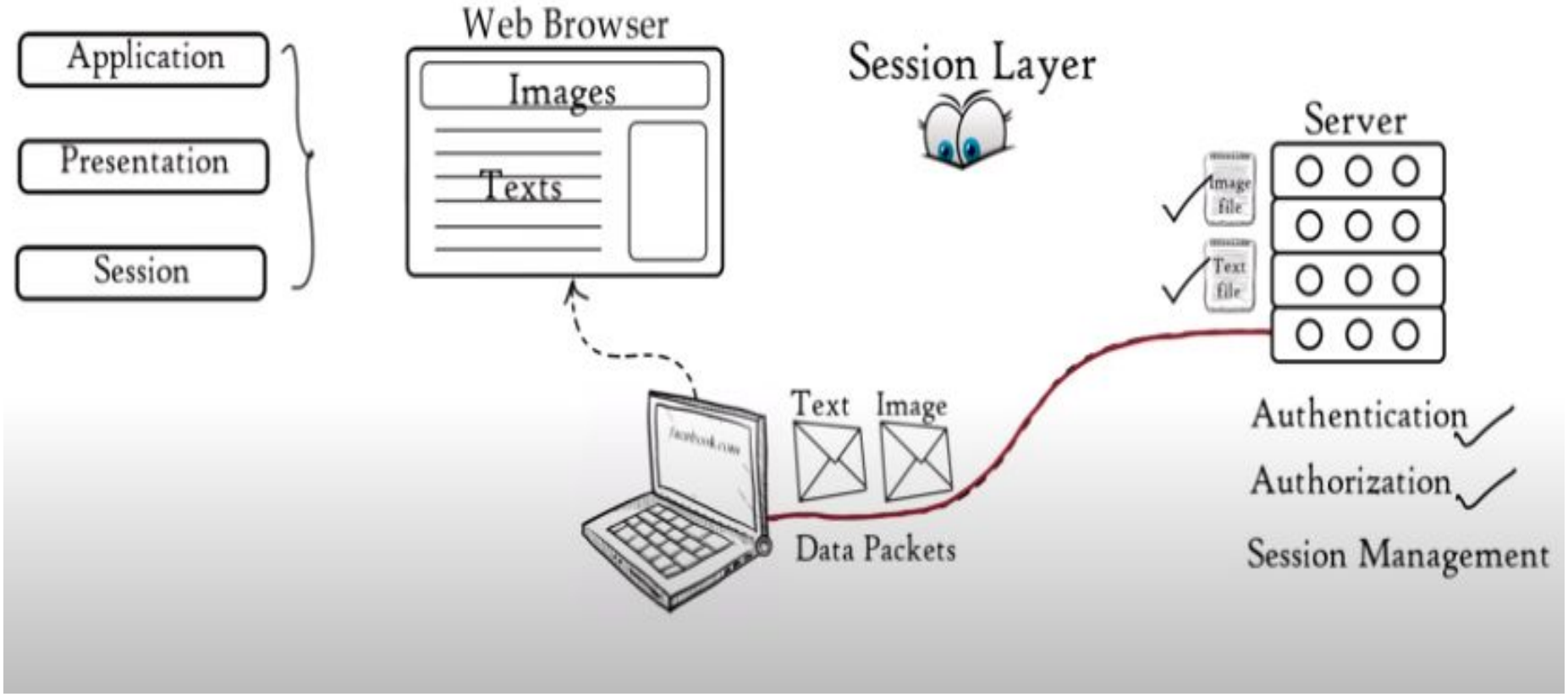


FTP HTTP/S SMTP Telnet

# Presentation Layer



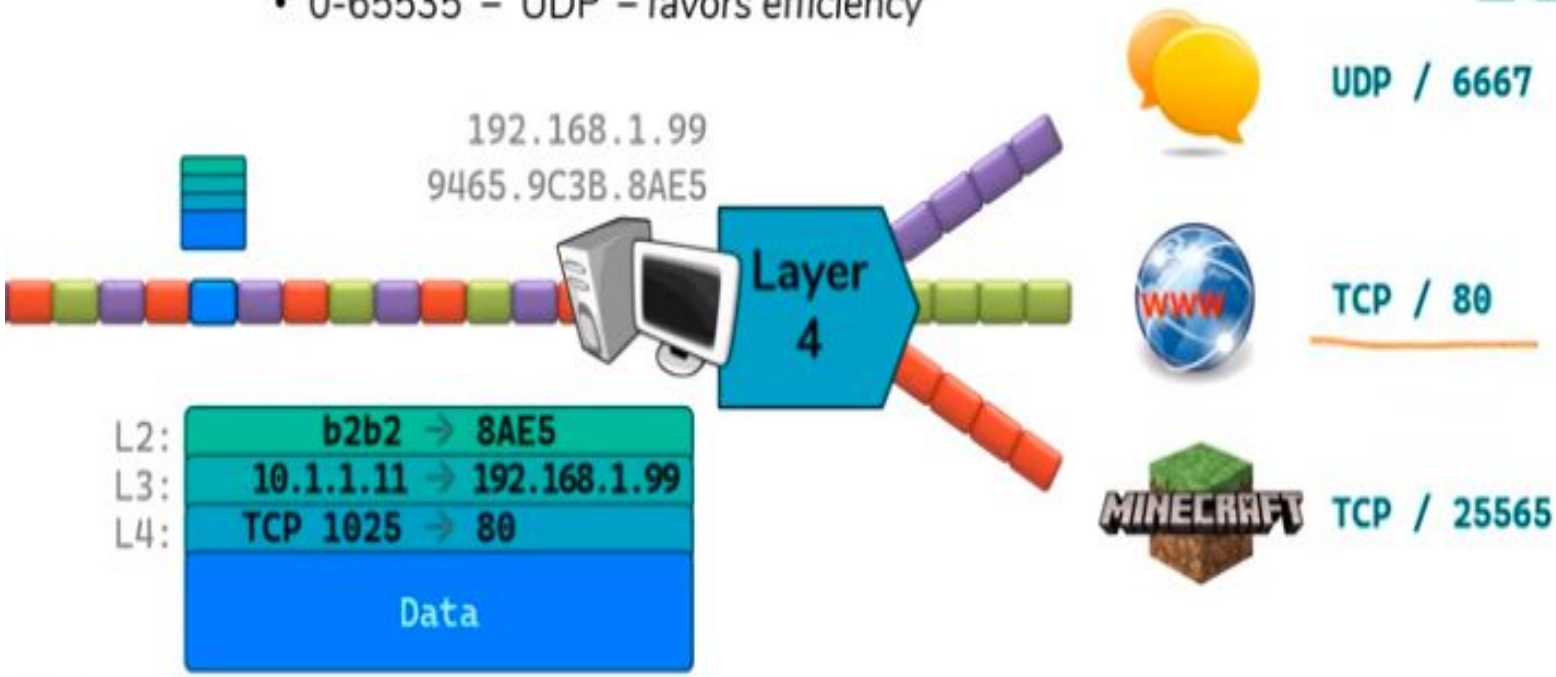

OSI Model
7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical
Internet
Shop Now
@#$%0&1!@#$%0
Converting
Encrypted
Shop Now
10101100
!@#$%0&1
REALPARS


Application
Presentation
Session
Web Browser
Images
Texts
Session Layer
Server
Image file
Text file
Text
Image
Data Packets
Authentication
Authorization
Session Management

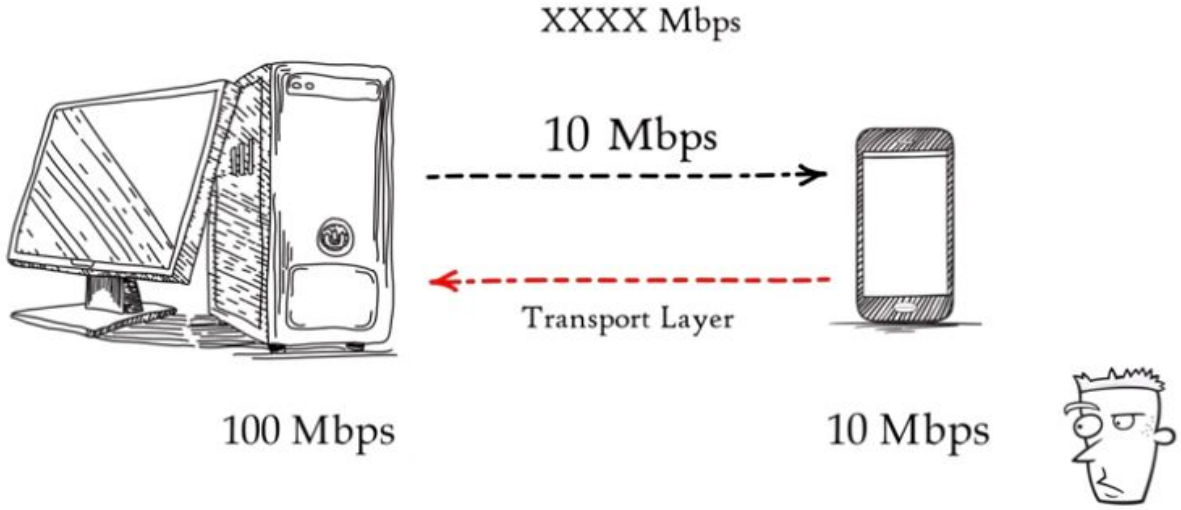
# OSI Model

- **Layer 4 – Transport – Service to Service**
  - Distinguish data streams
  - Addressing Scheme – Ports
    - 0-65535 – TCP – *favors reliability*
    - 0-65535 – UDP – *favors efficiency*

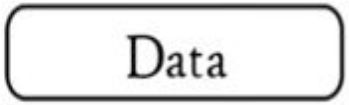
| | |
|---|---|
| 7 | Application |
| 6 | Presentation |
| 5 | Session |
| **4** | **Transport** |
| 3 | Network |
| 2 | Data Link |
| 1 | Physical |



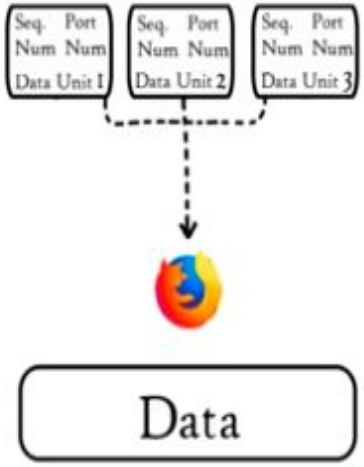
## Flow Control:



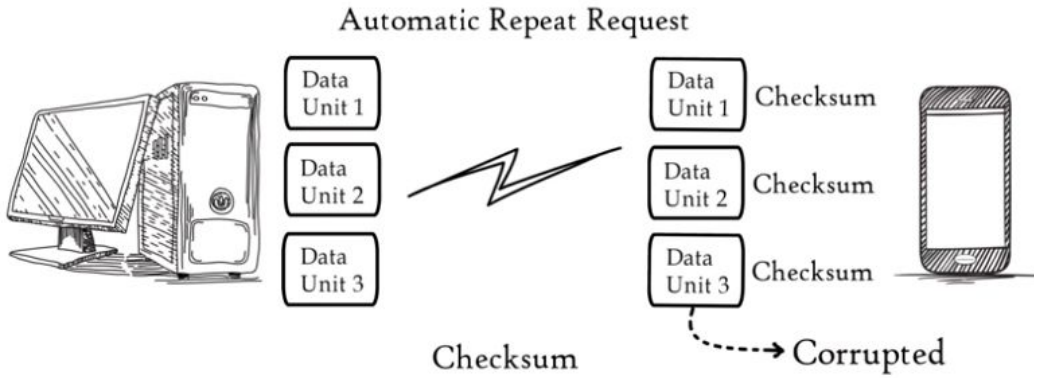

## Segmentation:




Segments




## Error Control:



# Transport Layer

| Services: | | Protocols: |
|---|---|---|
| - Connection-oriented Transmission | ----------> | Transmission Control Protocol (TCP) |
| - Connectionless Transmission | ----------> | User Datagram Protocol (UDP) |

## UDP



No feedback

## TCP

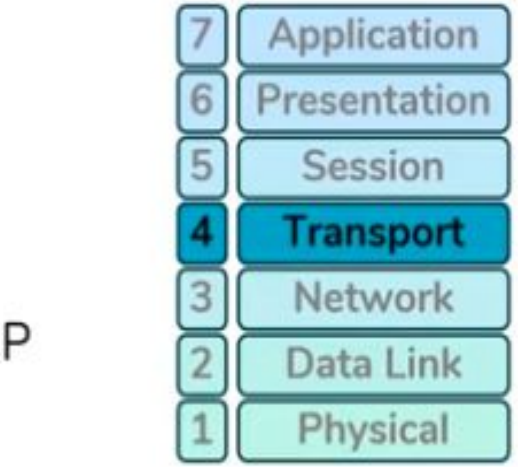


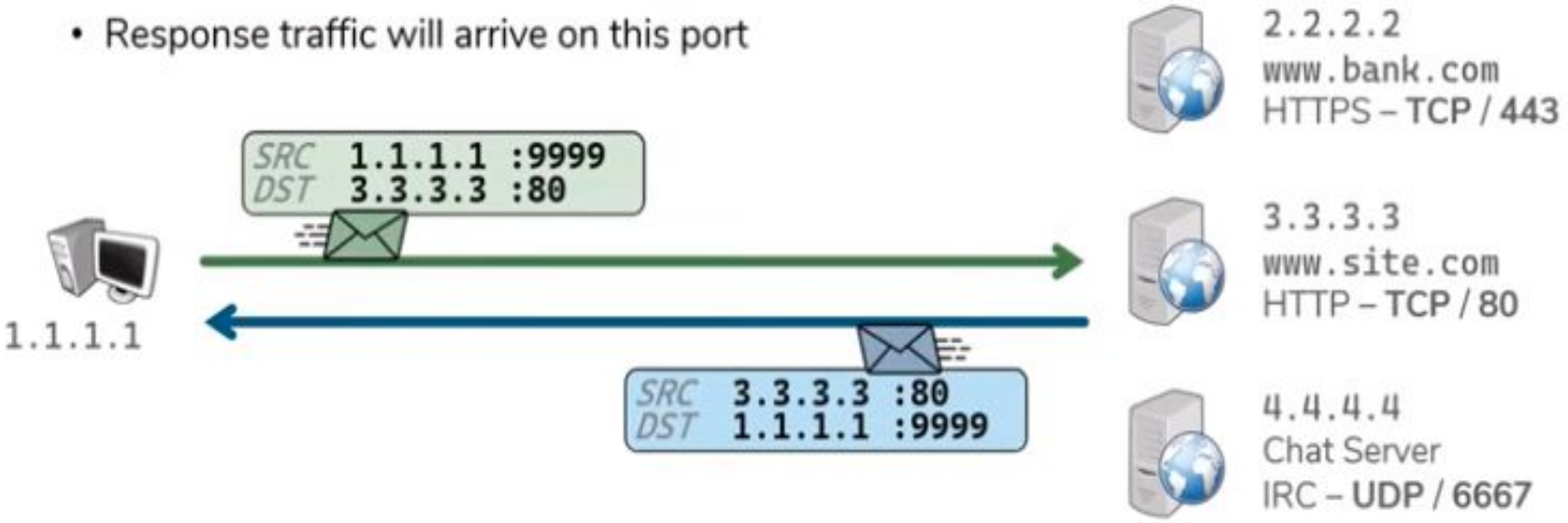
Feedback



# OSI Model

- **Layer 4 – Transport – Service to Service**
  - Distinguish data streams
  - Addressing Scheme – Ports – 0-65535, TCP or UDP
  - Servers listen for requests to pre-defined Ports
  - Clients select random Port for each connection
    - Response traffic will arrive on this port

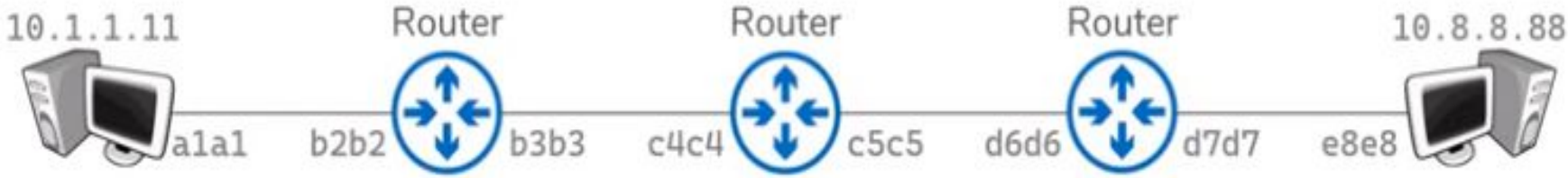


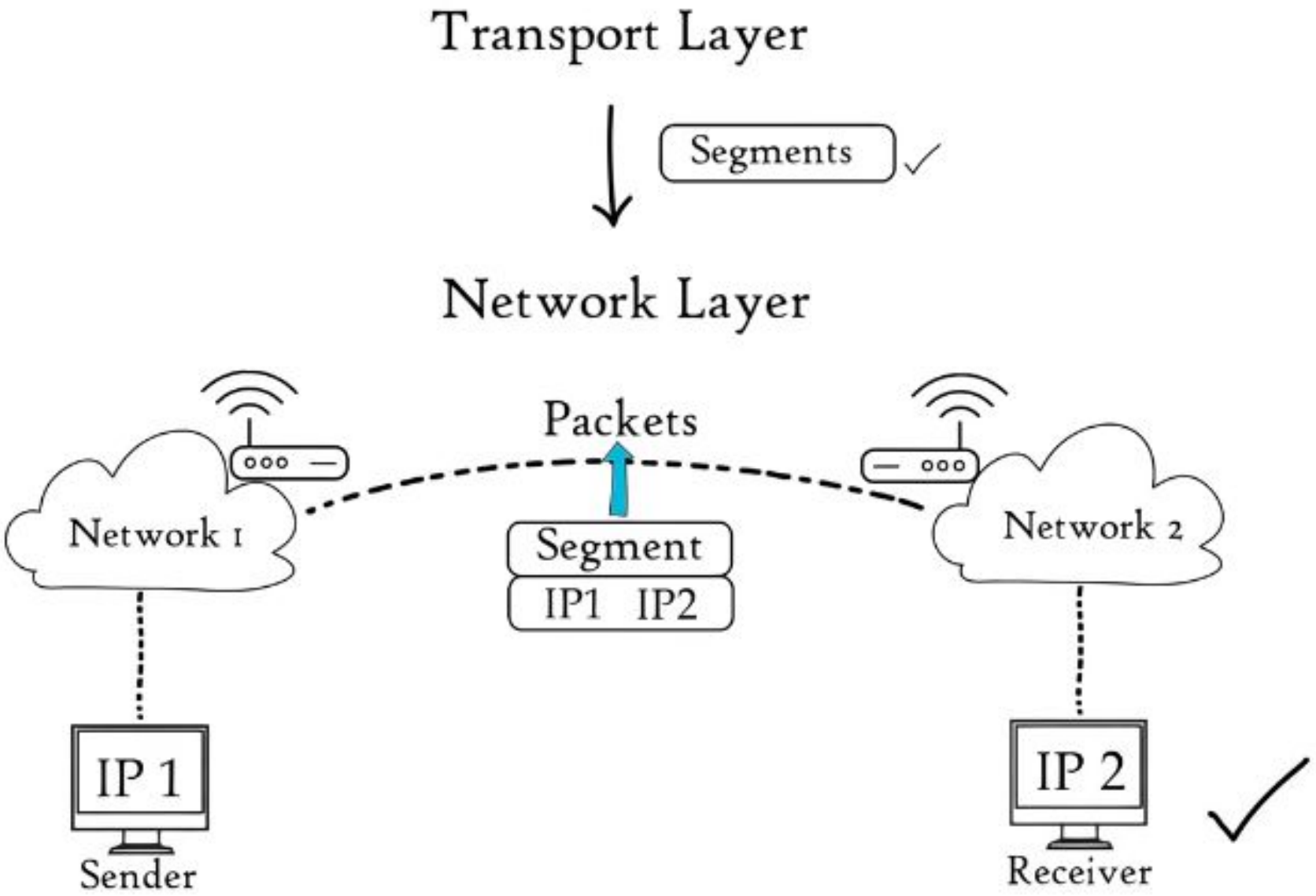



# OSI Model

- **Layer 3 – Network – End to End**
  - Addressing Scheme – IP addresses
    - 32 bits, represented as 4 octets, each 0-255
  - L3 Technologies: Routers, Hosts, *(anything with an IP)*




Transport Layer
Segments
Network Layer
Packets
Network 1
Segment
IP1 IP2
Network 2
IP 1
Sender
IP 2
Receiver

# Network layer

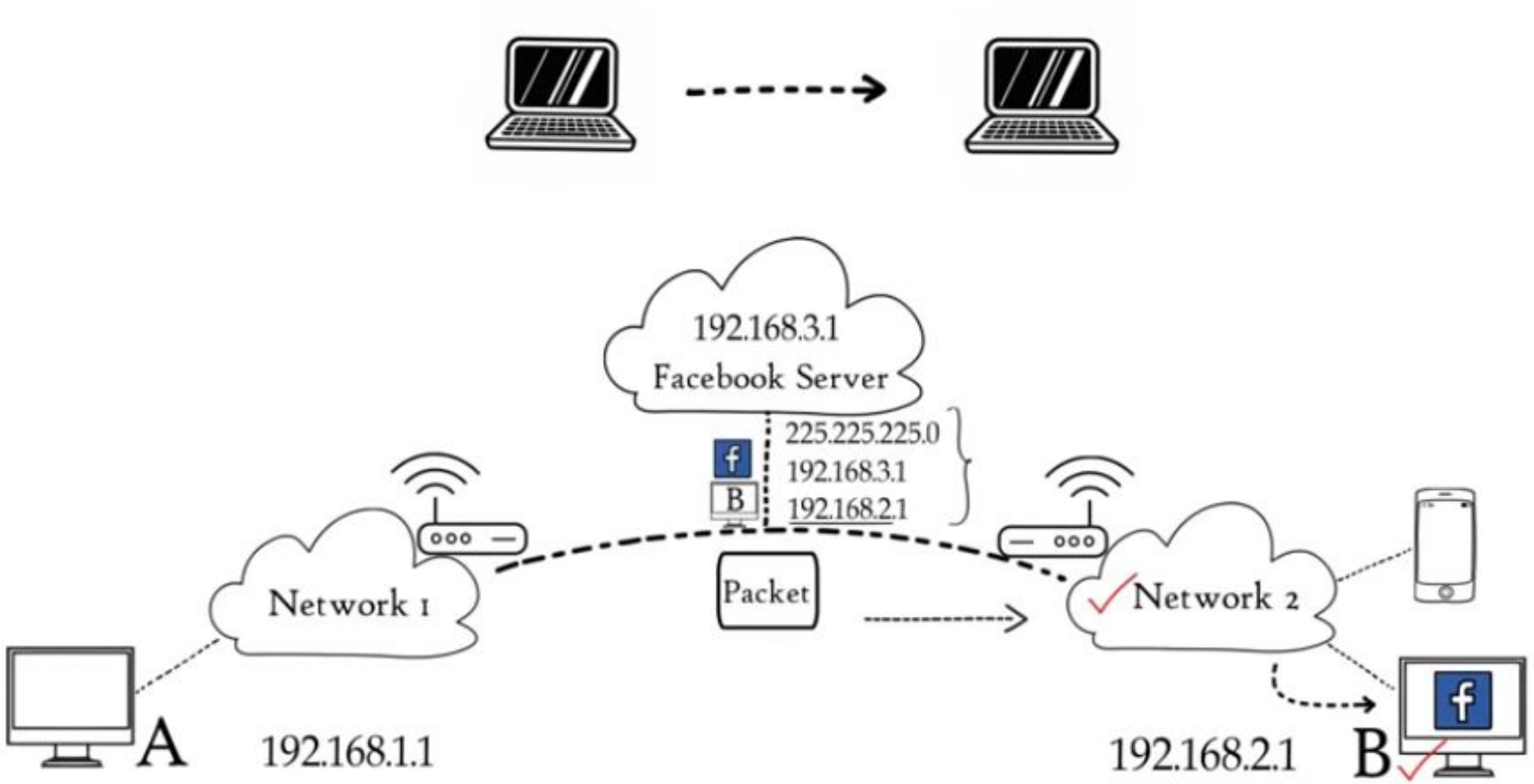
Logical Addressing



Path determination

Routing

# Routing


192.168.3.1
Facebook Server
225.225.225.0
192.168.3.1
192.168.2.1
B
Network 1
Packet
Network 2
A
192.168.1.1
192.168.2.1
B

# Path determination

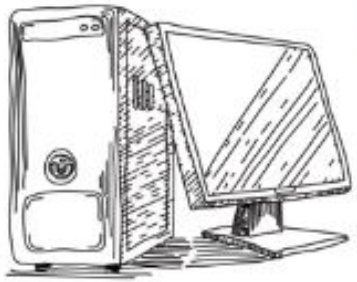
OSPF

BGP

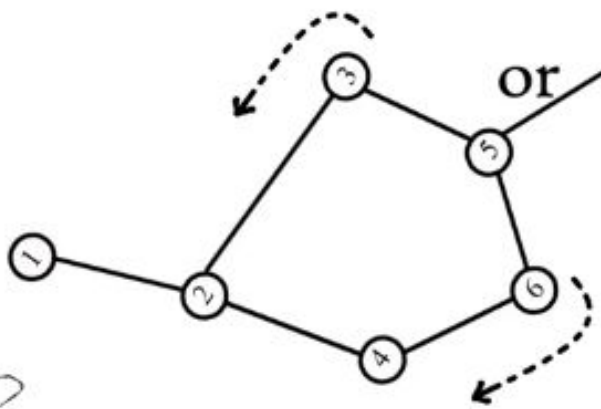
IS-IS

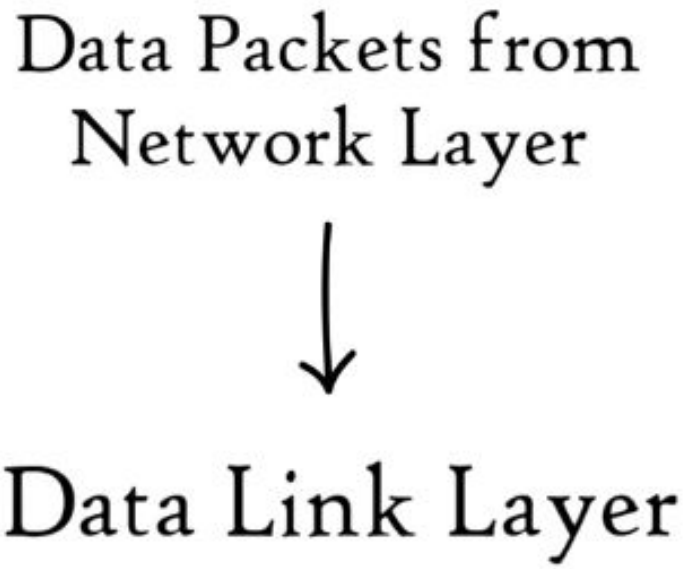


Receiver



Sender

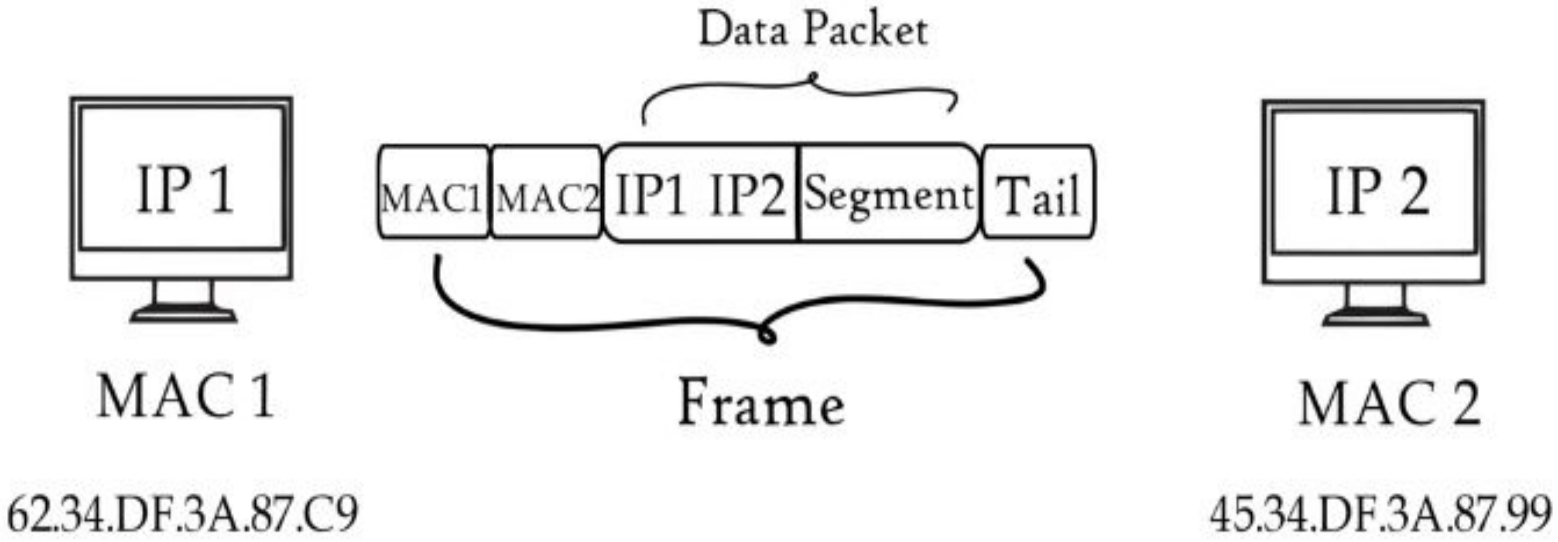


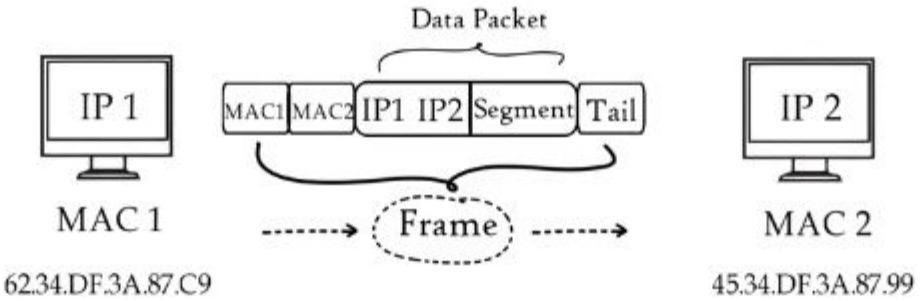

Data Packets from
Network Layer
Data Link Layer

# Data Link Layer

- Logical addressing : Network layer

- Physical addressing : Data Link layer




Data Packet
IP 1
MAC1
MAC2
IP1
IP2
Segment
Tail
IP 2
MAC 1
Frame
MAC 2
62.34.DF.3A.87.C9
45.34.DF.3A.87.99


Network 1
NIC
Media
NIC
Network 2



# Data Link Layer

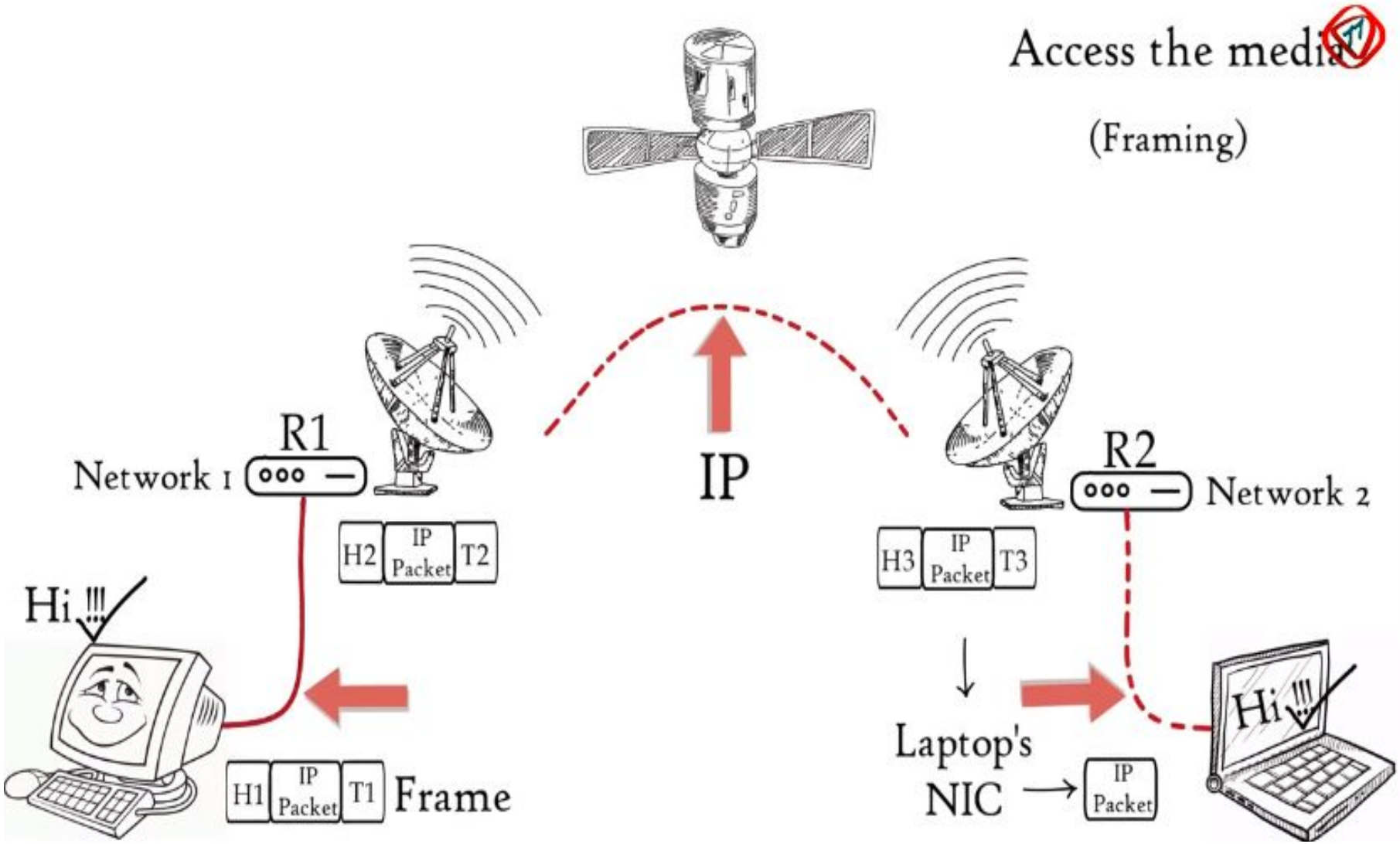


Access the media

(Framing)

Controls how data is placed and received from the media

(Media Access Control)
(Error Detection)


Access the media
(Framing)
IP
R1
Network 1
R2
Network 2
H2
IP Packet
T2
H3
IP Packet
T3
Hi
H1
IP Packet
T1
Frame
Laptop's NIC
IP Packet
Hi

Controls how data is placed and received from the media

(Media Access Control)

(Error Detection)

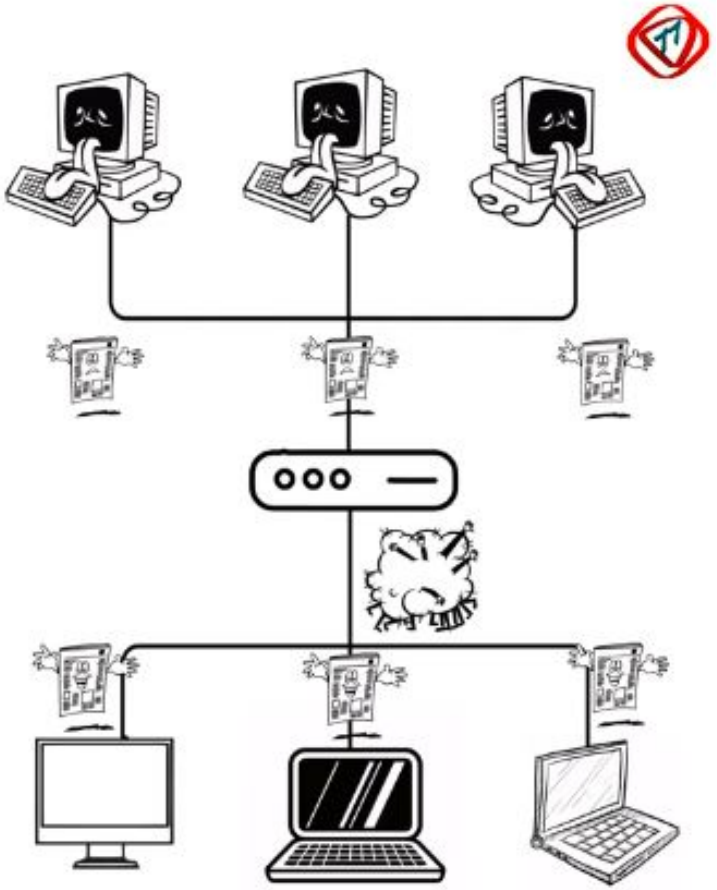



DATA LINK LAYER

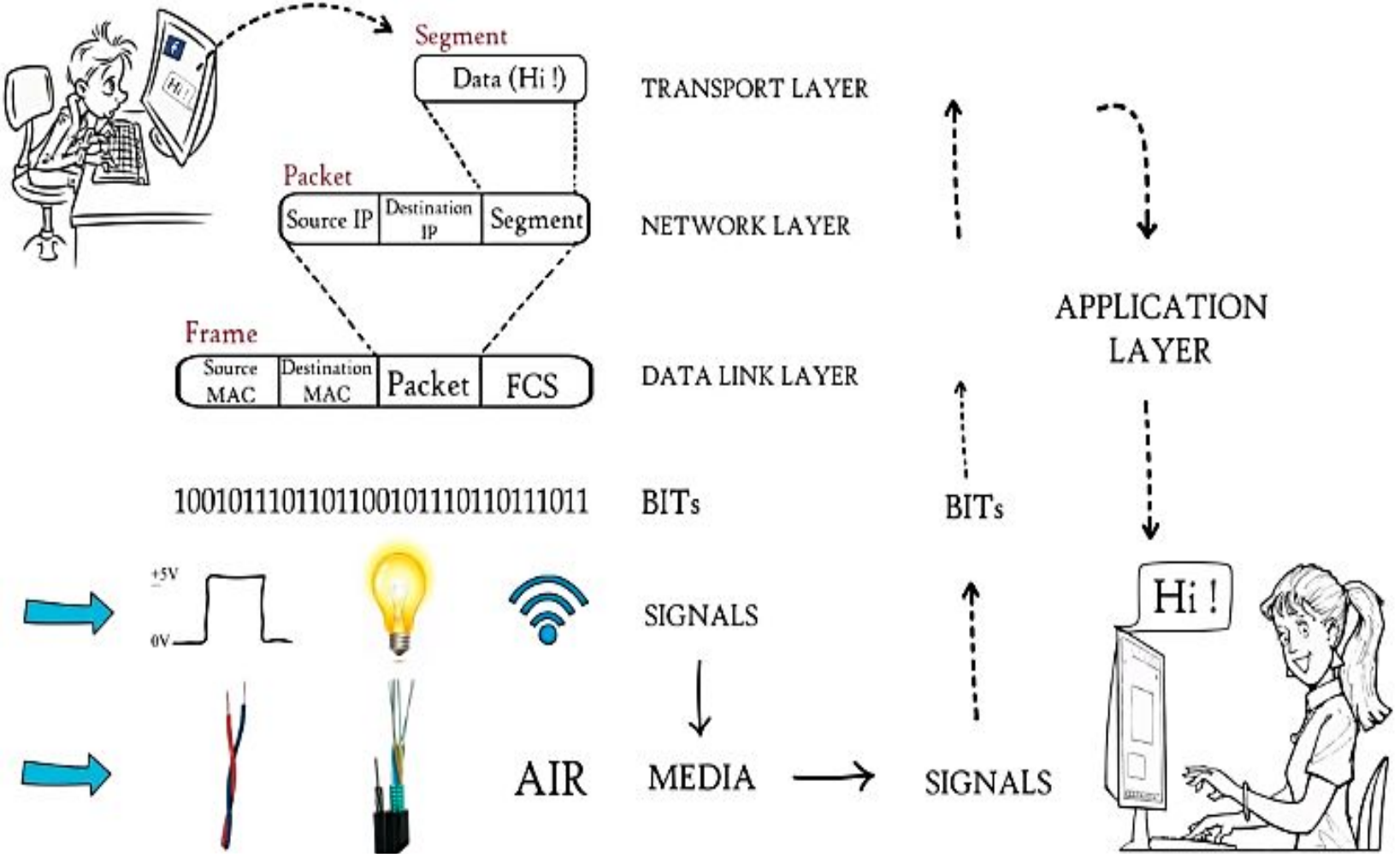


CSMA

LAN Cable, Optical Fiber, Air



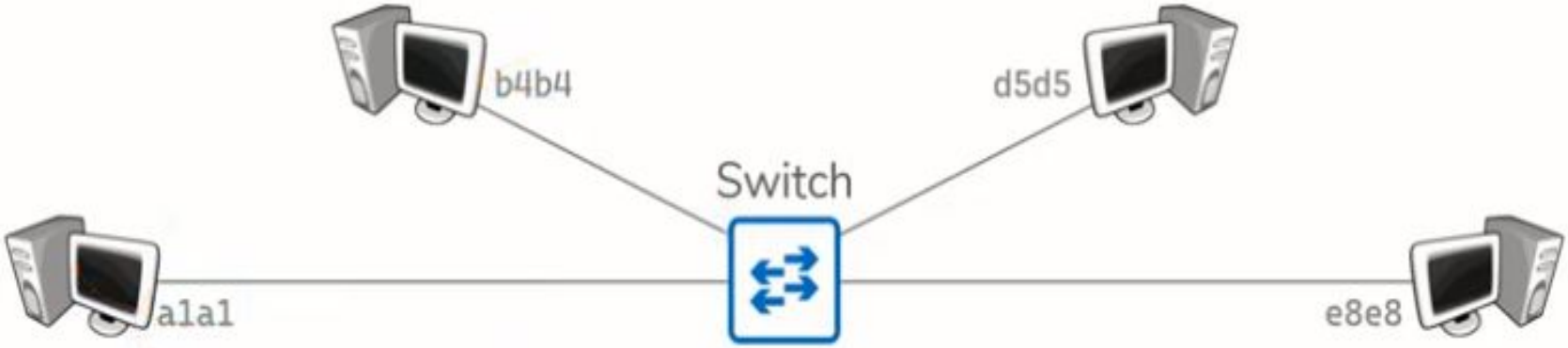
# Physical Layer



# OSI Model

- **Layer 2 – Data Link – Hop to Hop**
  - Interacts with the Wire (i.e., Physical layer)
  - Addressing Scheme – MAC addresses
  - L2 Technologies: NICs, Switches

| | |
|---|---|
| 7 | Application |
| 6 | Presentation |
| 5 | Session |
| 4 | Transport |
| 3 | Network |
| **2** | **Data Link** |
| 1 | Physical |



# OSI Model

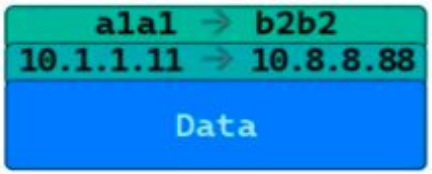
- Layer 3 – IP Addresses – End to End delivery
- Layer 2 – MAC Addresses – Hop to Hop delivery

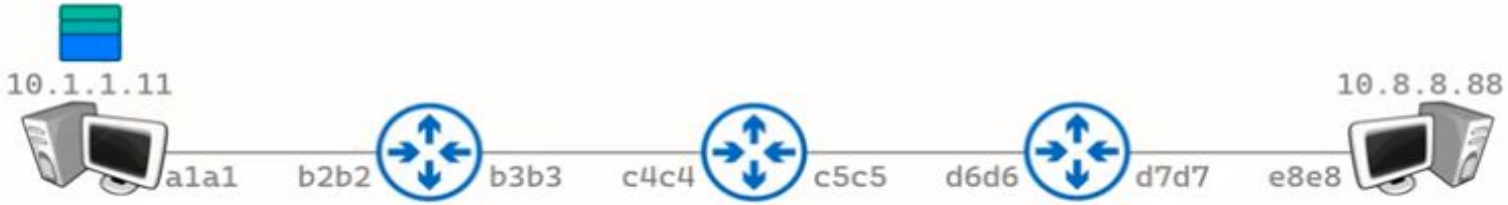








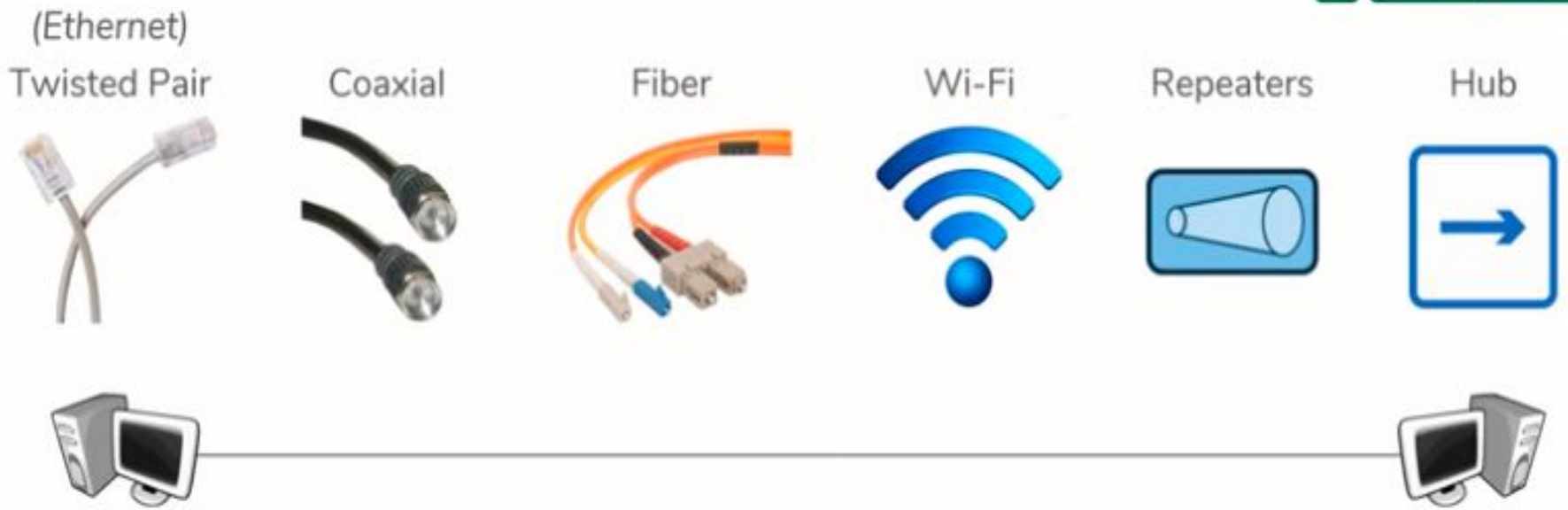



# OSI Model



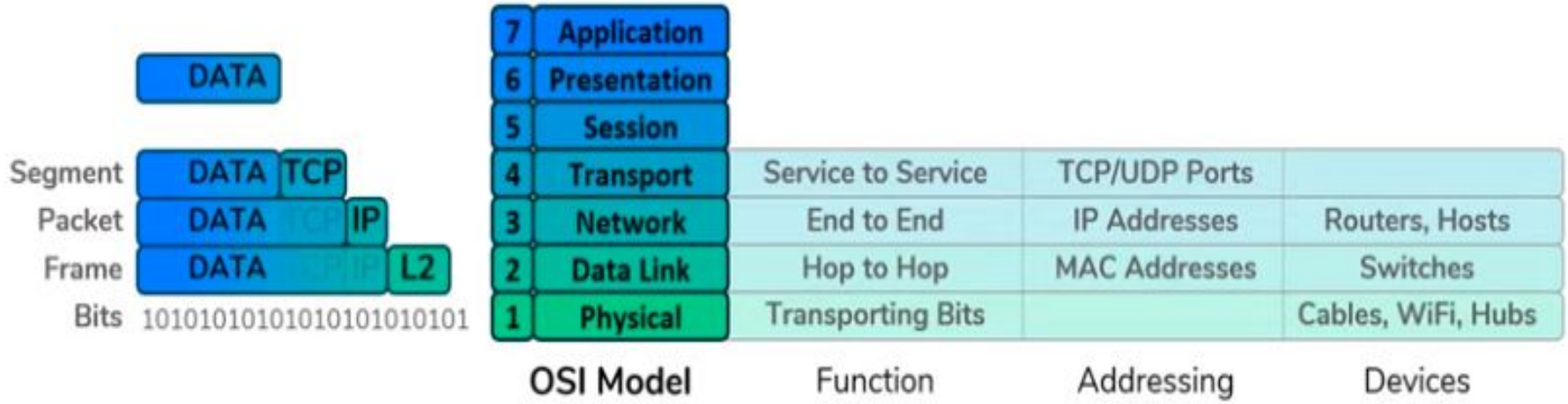

- **Layer 1 – Physical – Transporting Bits**
  - Computer data exists in the form of Bits (1's and 0's)
  - Something has to transport those bits between hosts
  - L1 Technologies: Cables, Wifi, Repeaters, Hubs



# OSI Model

- Network Devices operate at specific layers
- Network Protocols operate at specific layers

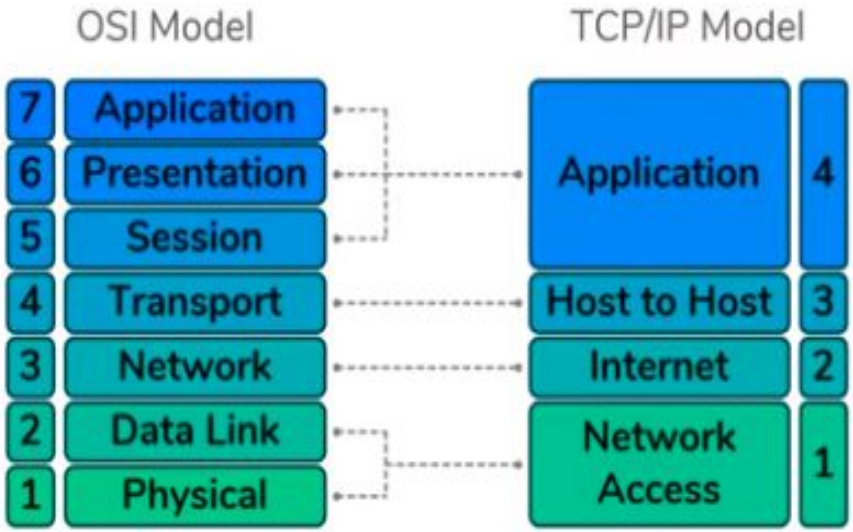
- **Neither of these are strict rules** – exceptions exist
- **OSI Model is simply a model** – not rigid rules everything adheres to
  - Conceptualization of what is required for data to flow through the Internet



# OSI Model

- **Layer 5, 6, 7 – Session, Presentation, Application**
  - Distinction between these layers is somewhat vague
  - Other Networking Models combine these into one layer
  - L1-L4 are most important to understand how data flows



## 1) The Physical Layer

The Physical Layer is the bottom most layer and is associated with electrical, mechanical and functional aspects of the transmission media for information and receiving over internet.

## 2) The Data Link Layer

The Data Link Layer is second from bottom and comes under the lower layer category. It ensures that the data must be synchronized, error detection and control are enabled.

## 3) The Network Layer

The Network Layer is third from bottom in OSI model and is responsible for establishing data communication channel between multiple networks or devices or hosts or nodes.

## E) The Transport Layer

Transport Layer is the middle most layer in OSI model and it acts as Network Independent Layer. It has no idea about the functioning of lower layer i.e. physical, data link and network layers.

## E) The Session Layer

Session Layer is the fifth layer of OSI model and it provides appropriate sessions between users and entities, where user interacts. This layer can be used on the basis of resources available and it can be skipped too if not required.
For example : Login Sessions in online banking.

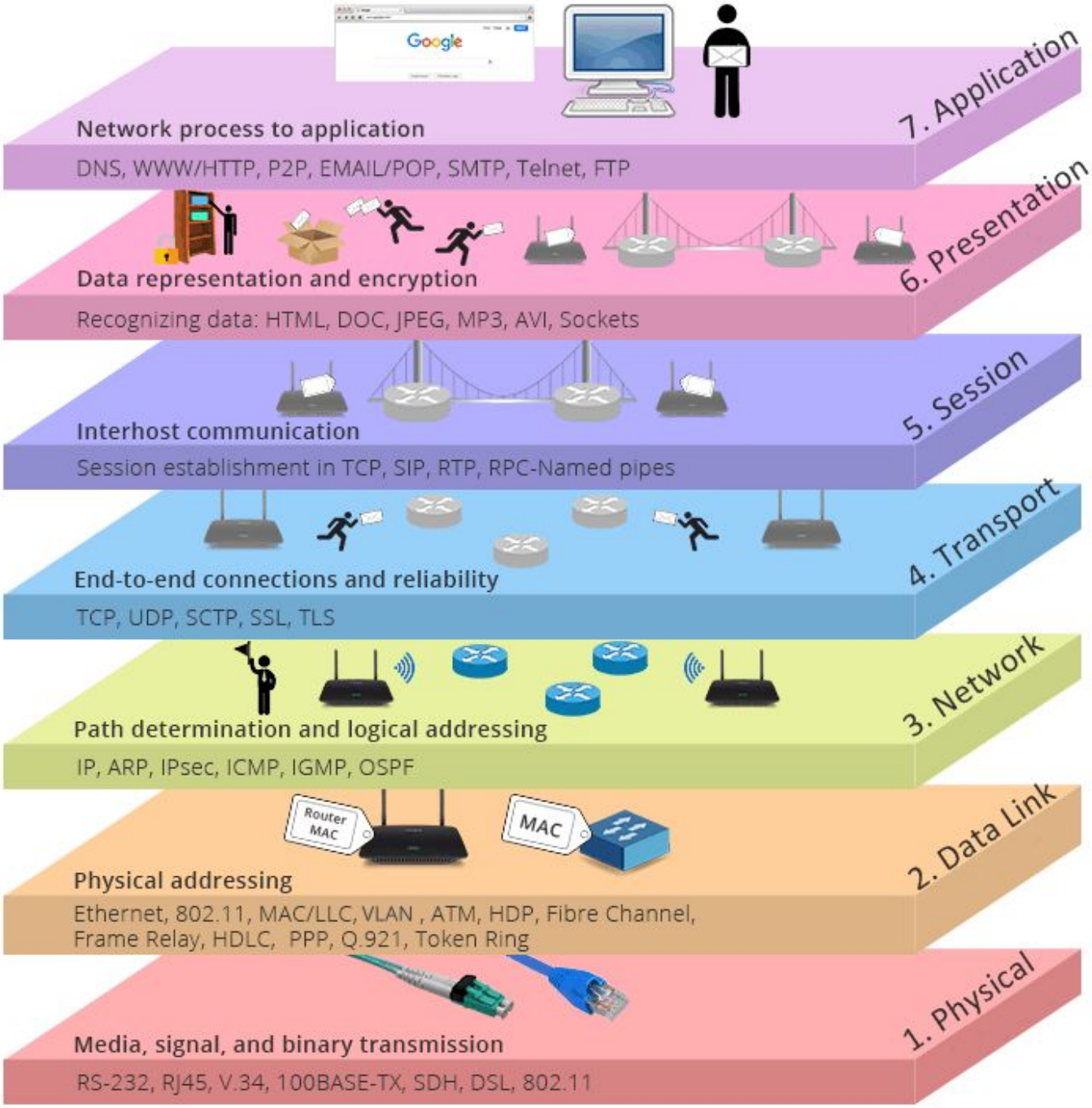
## F) The Presentation Layer

This is the sixth layer of OSI model and it provides appropriate representation of data through various data presentation techniques.

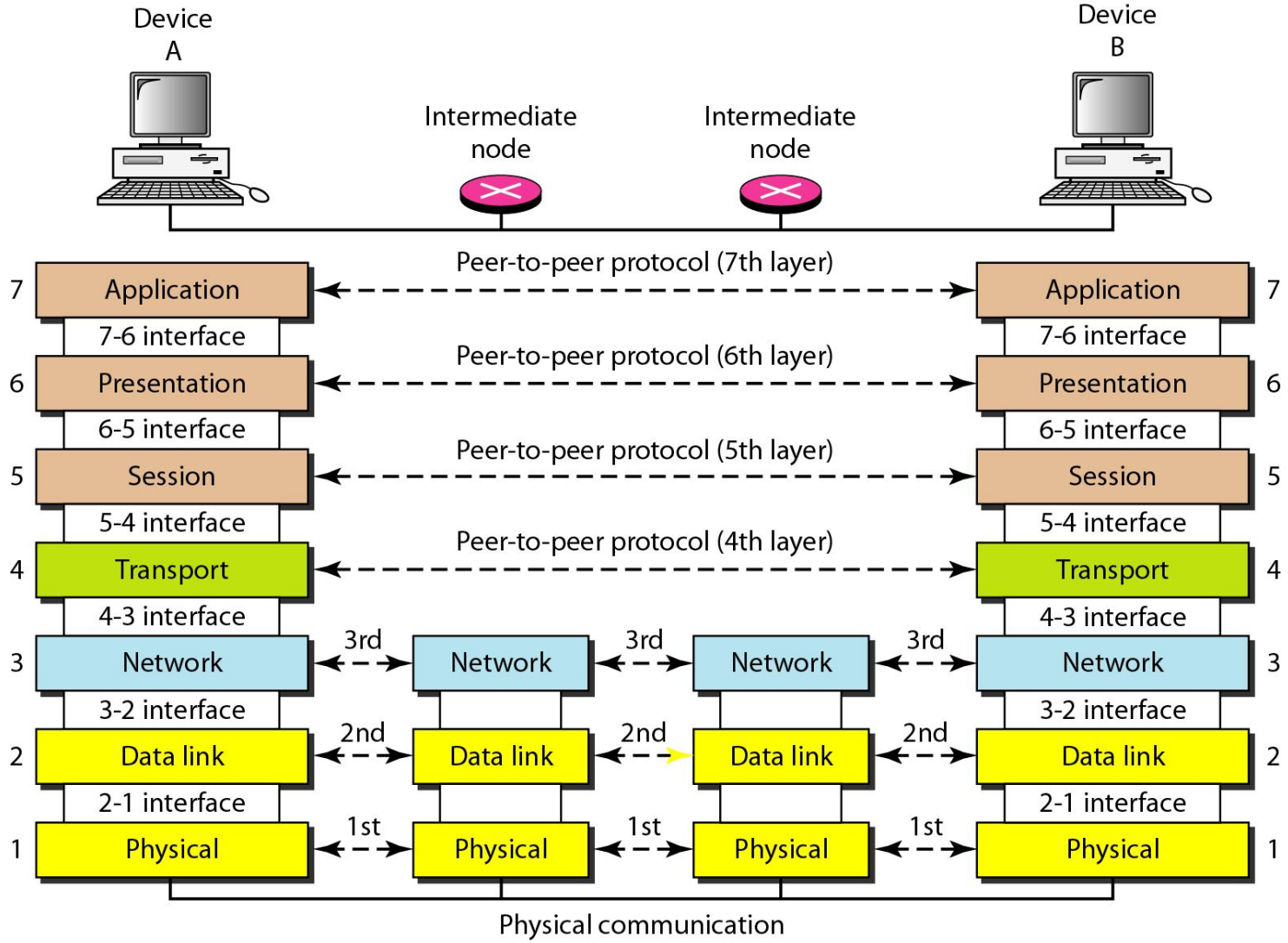
## G) The Application Layer

Application Layer is the topmost layer of the OSI model and has the responsibility for providing interface between various users and application.

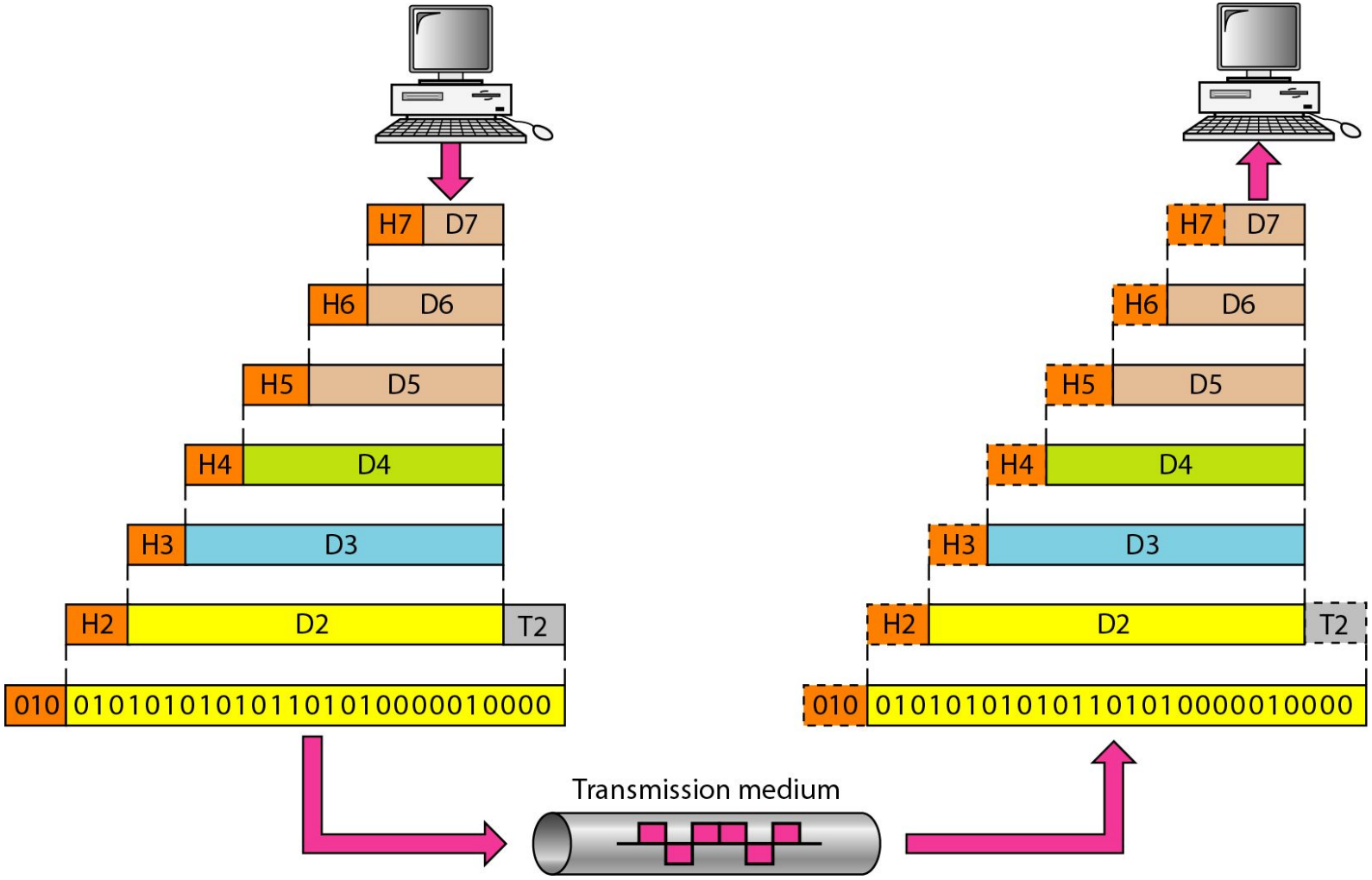
| | |
|---|---|
| 7. Application | • Provides a user interface |
| 6. Presentation | • Presents Data<br>• Handles encryption and decryption |
| 5. Session | • Maintains distinction between data of separate applications<br>• Provides dialog control between hosts |
| 4. Transport | • Provides End-to-End connections<br>• Provides reliable or unreliable delivery and flow control |
| 3. Network | • Provides Logical Addressing<br>• Provides Path determination using logical addressing |
| 2. Data Link | • Provides media access and physical addressing |
| 1. Physical | • Converts digital data so that it can be sent over the physical medium<br>• Moves data between hosts |


Google
7. Application
Network process to application
DNS, WWW/HTTP, P2P, EMAIL/POP, SMTP, Telnet, FTP
6. Presentation
Data representation and encryption
Recognizing data: HTML, DOC, JPEG, MP3, AVI, Sockets
5. Session
Interhost communication
Session establishment in TCP, SIP, RTP, RPC-Named pipes
4. Transport
End-to-end connections and reliability
TCP, UDP, SCTP, SSL, TLS
3. Network
Path determination and logical addressing
IP, ARP, IPsec, ICMP, IGMP, OSPF
Router MAC
MAC
2. Data Link
Physical addressing
Ethernet, 802.11, MAC/LLC, VLAN , ATM, HDP, Fibre Channel,
Frame Relay, HDLC, PPP, Q.921, Token Ring
1. Physical
Media, signal, and binary transmission
RS-232, RJ45, V.34, 100BASE-TX, SDH, DSL, 802.11

## Figure 2.3 *The interaction between layers in the OSI model*



## Figure 2.4 *An exchange using the OSI model*



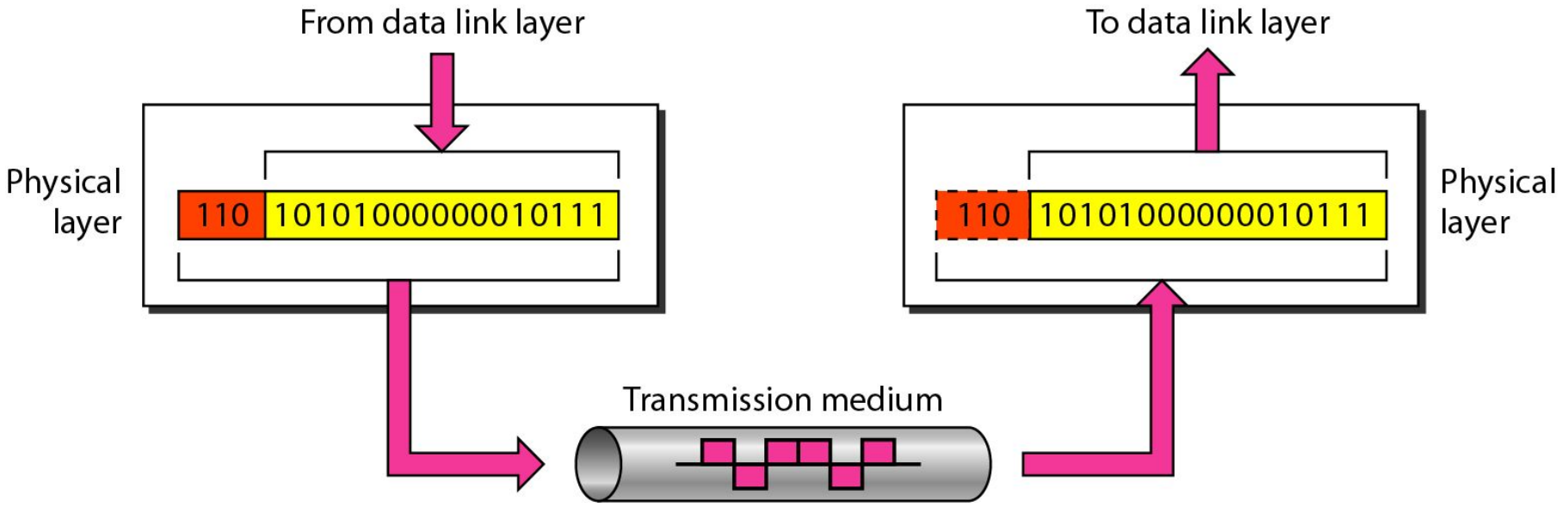
# 2-3 LAYERS IN THE OSI MODEL

*In this section we briefly describe the functions of each layer in the OSI model.*

***Topics discussed in this section:***

**Physical Layer**
**Data Link Layer**
**Network Layer**
**Transport Layer**
**Session Layer**
**Presentation Layer**
**Application Layer**

## Figure 2.5 *Physical layer*

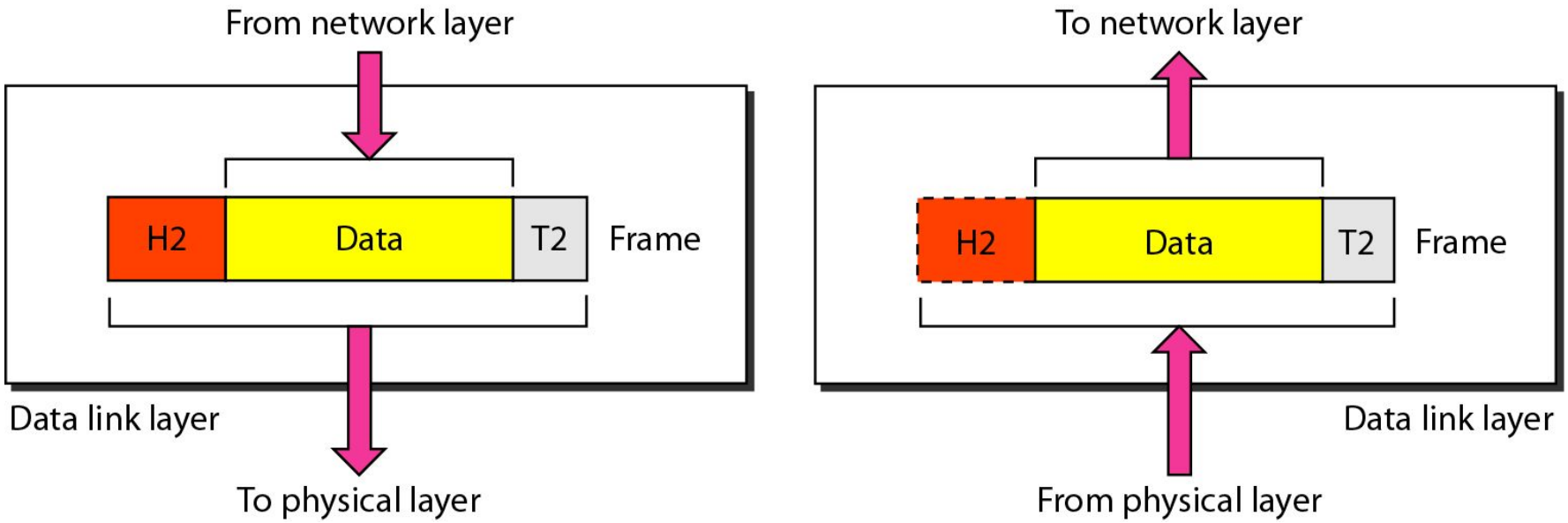




*Note*

**The physical layer is responsible for movements of individual bits from one hop (node) to the next.**

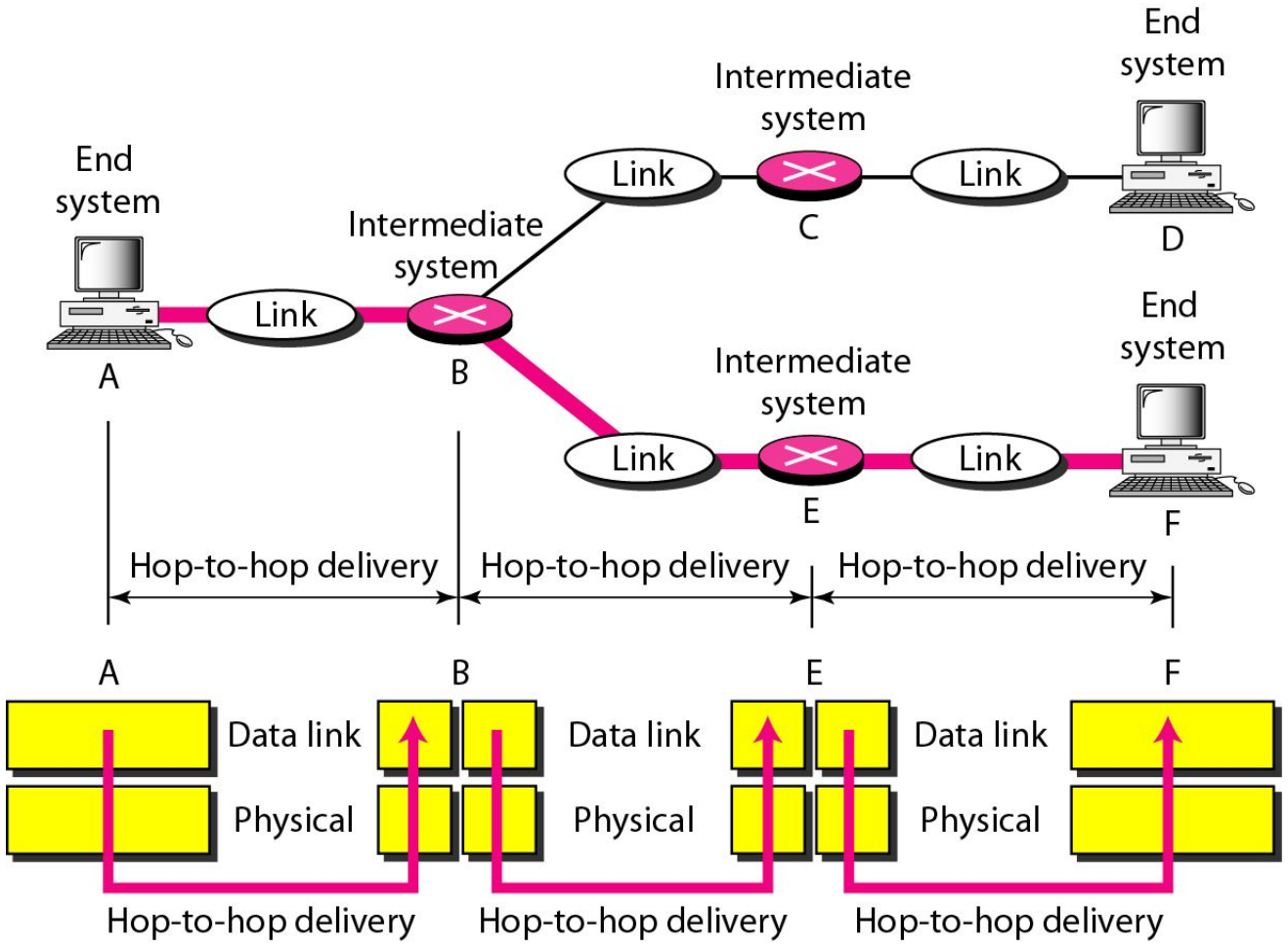
## Figure 2.6 *Data link layer*



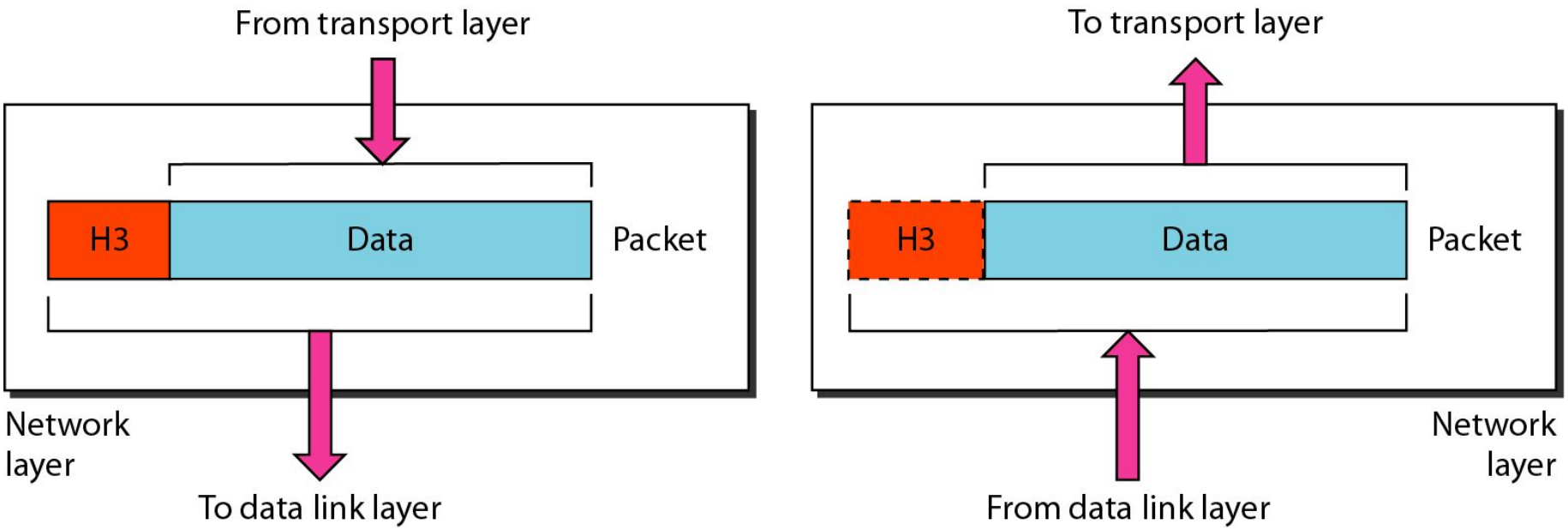
*Note*

**The data link layer is responsible for moving frames from one hop (node) to the next.**

# Figure 2.7 *Hop-to-hop delivery*



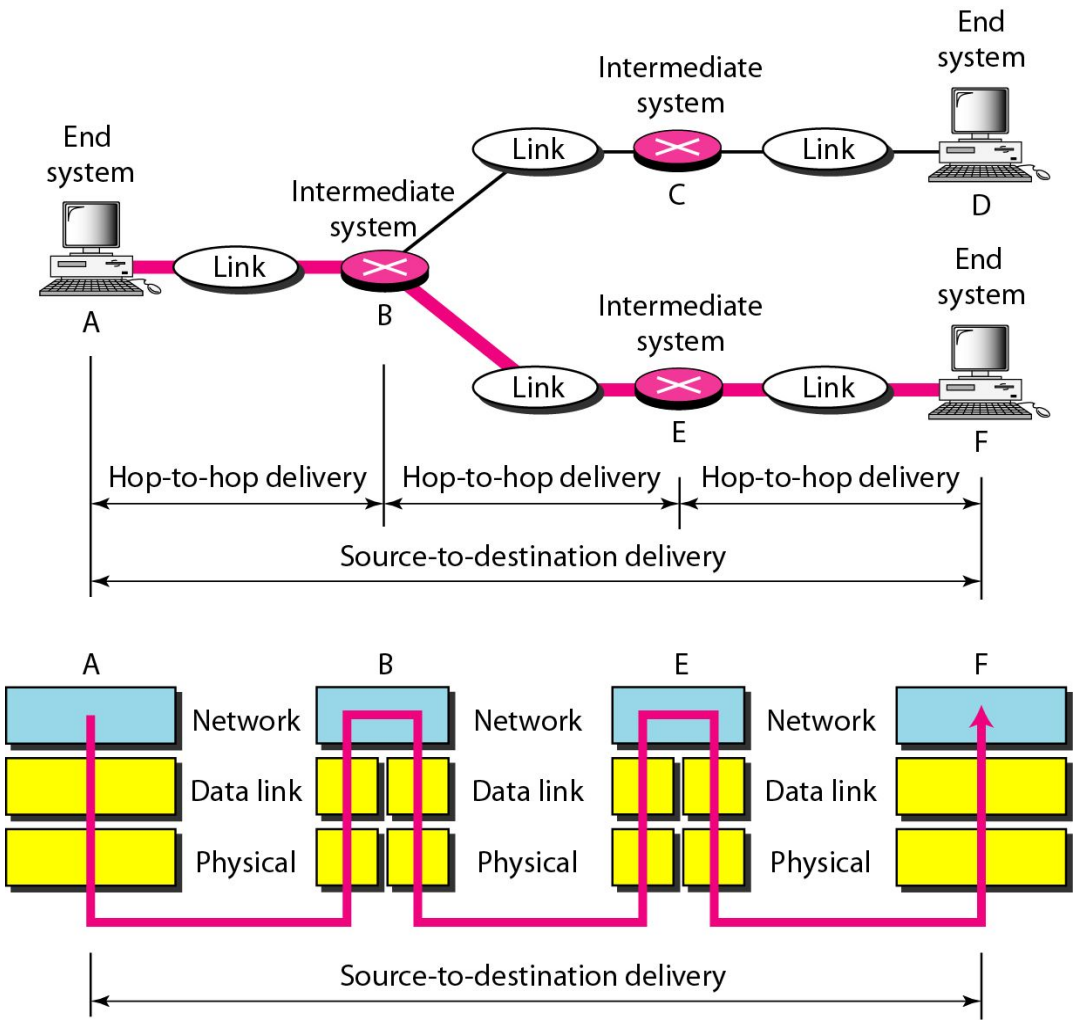
## Figure 2.8 *Network layer*



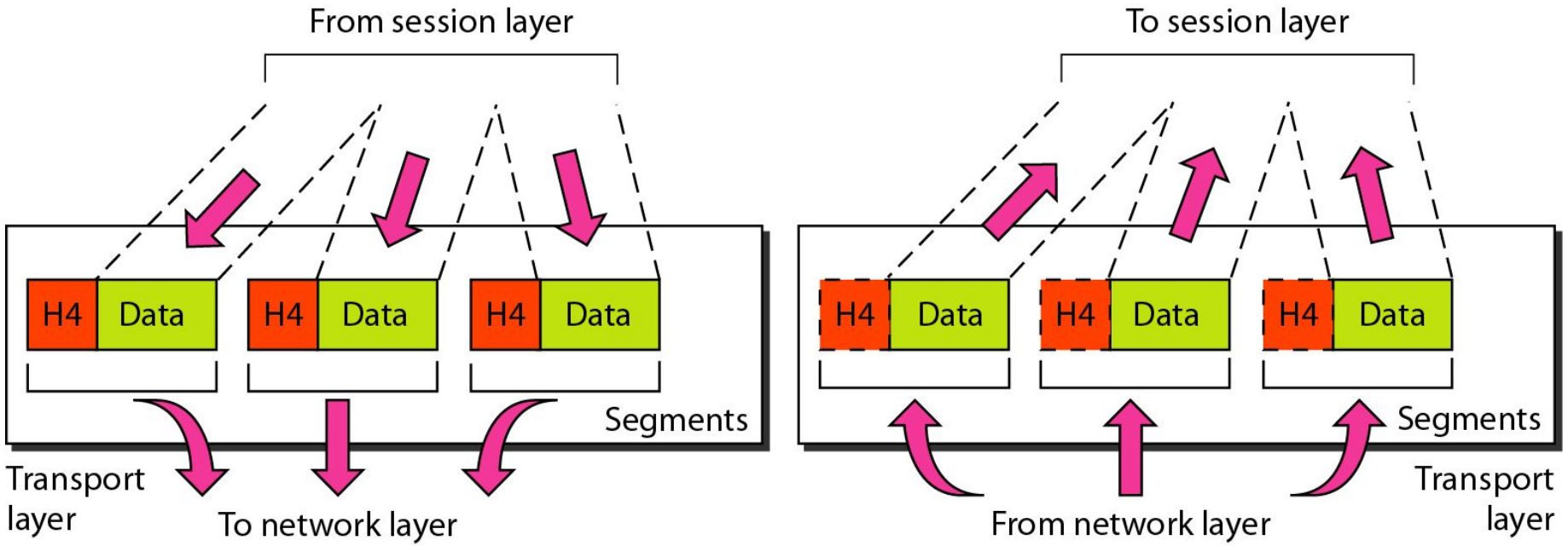
*Note*

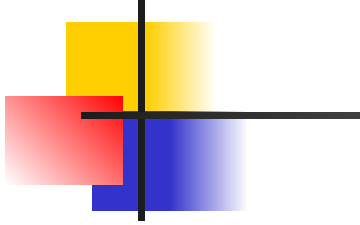
**The network layer is responsible for the delivery of individual packets from the source host to the destination host.**

# Figure 2.9 *Source-to-destination delivery*



## Figure 2.10 *Transport layer*

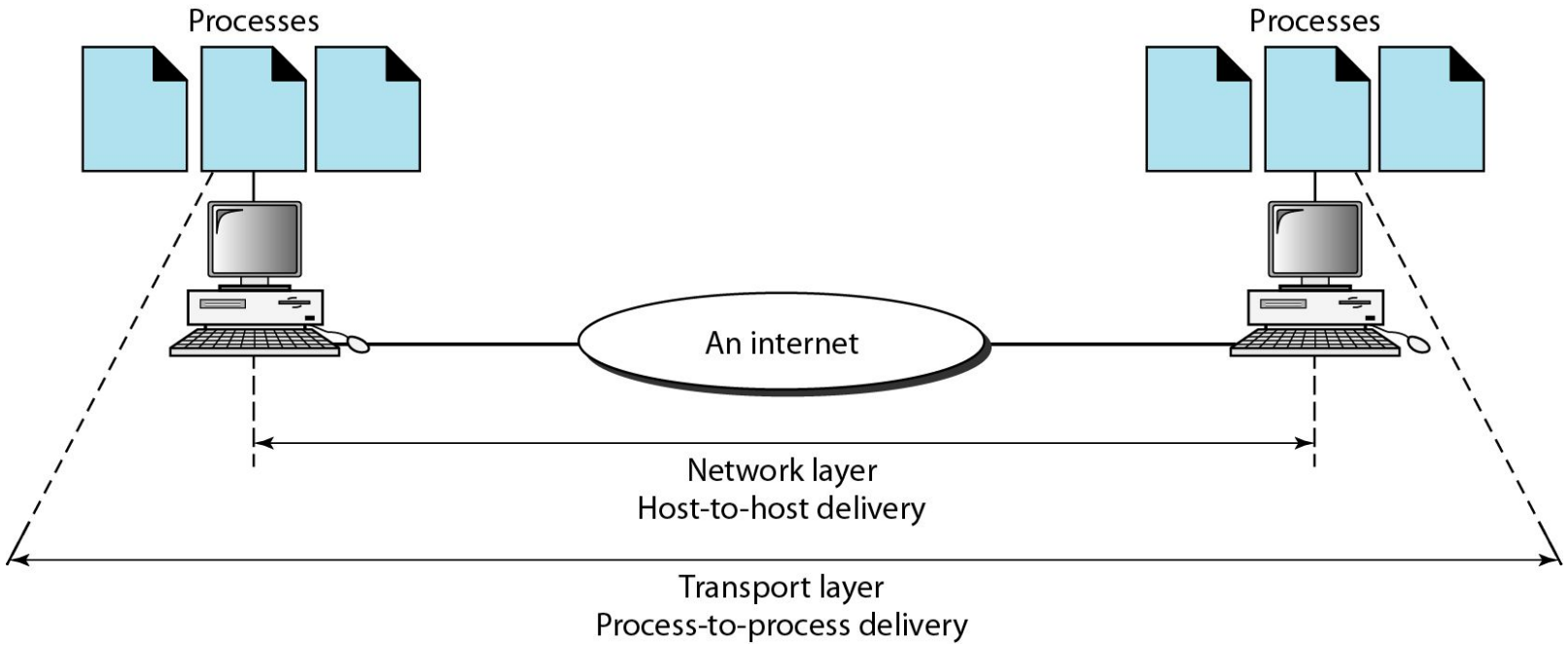




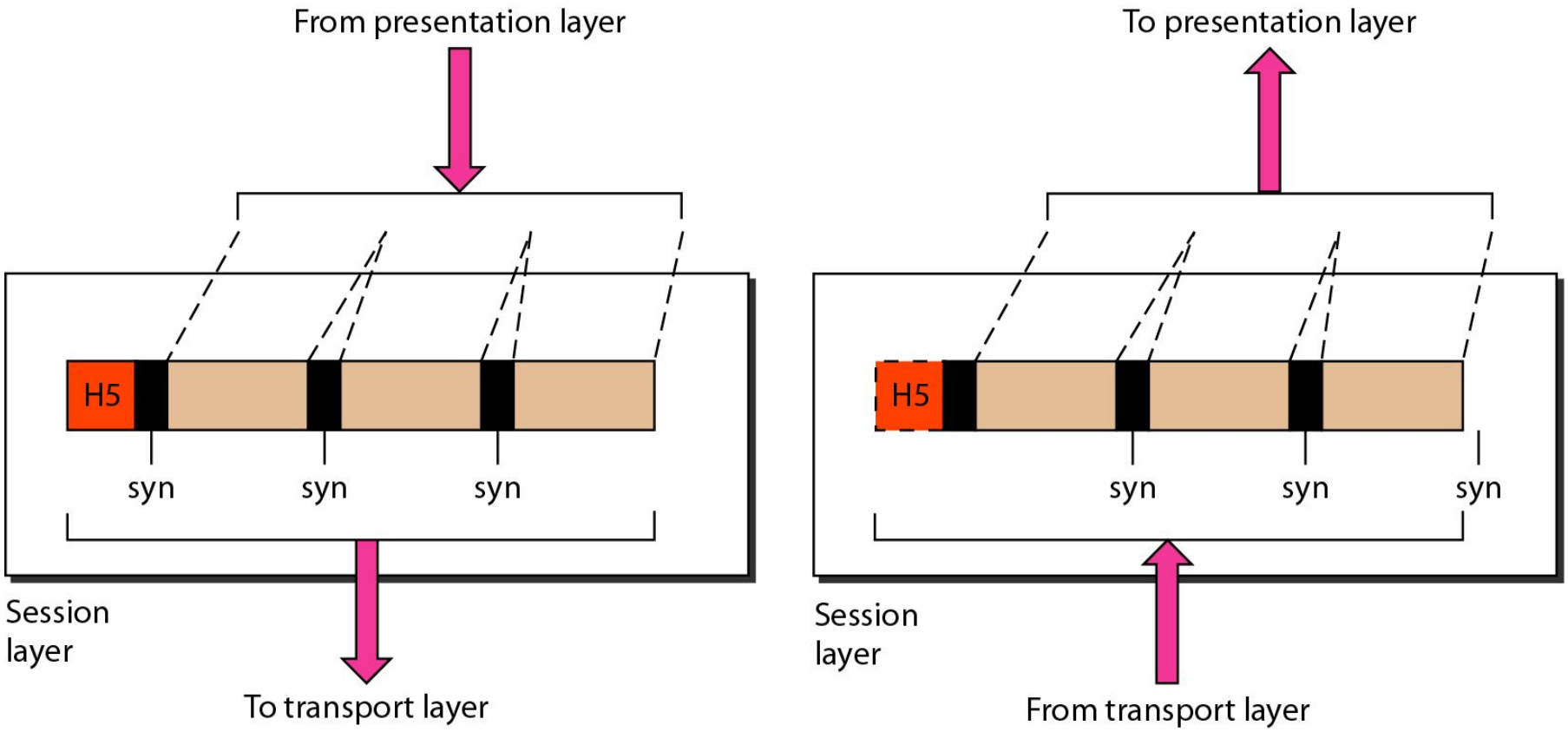
*Note*

**The transport layer is responsible for the delivery of a message from one process to another.**

## Figure 2.11 *Reliable process-to-process delivery of a message*



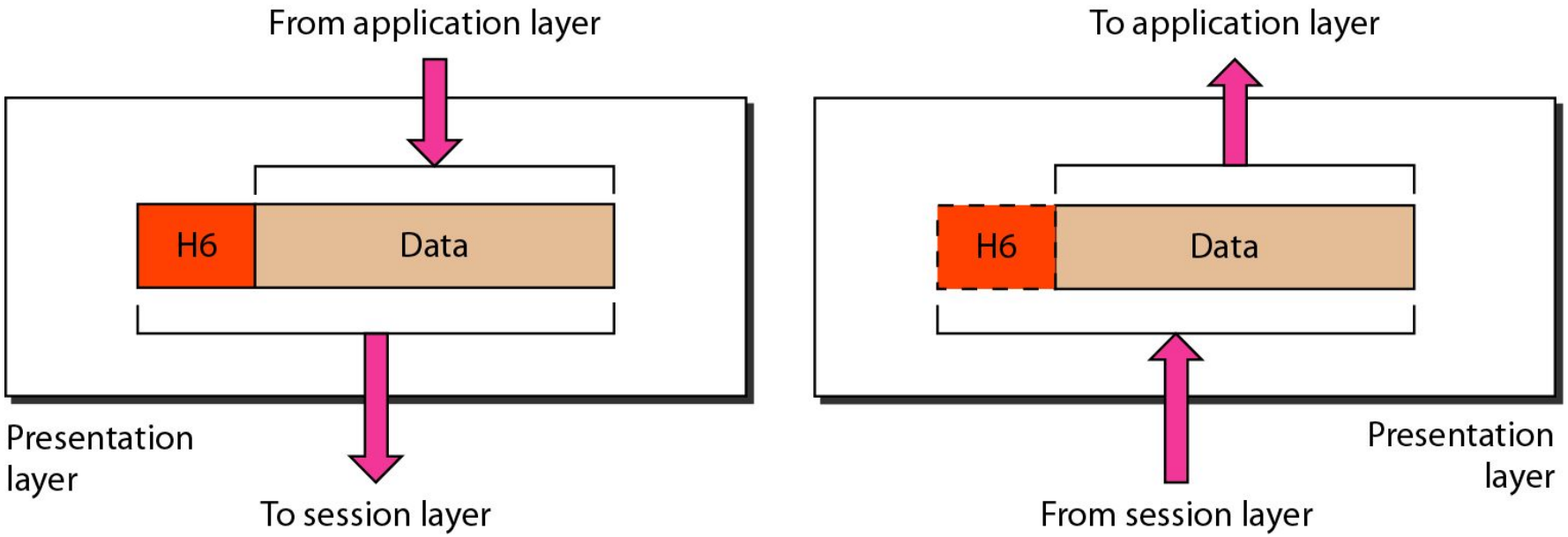
## Figure 2.12 *Session layer*

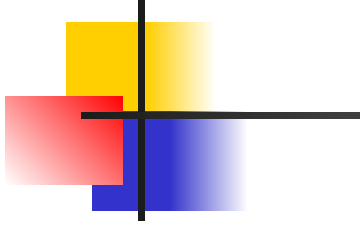


*Note*

**The session layer is responsible for dialog control and synchronization.**

## Figure 2.13 *Presentation layer*

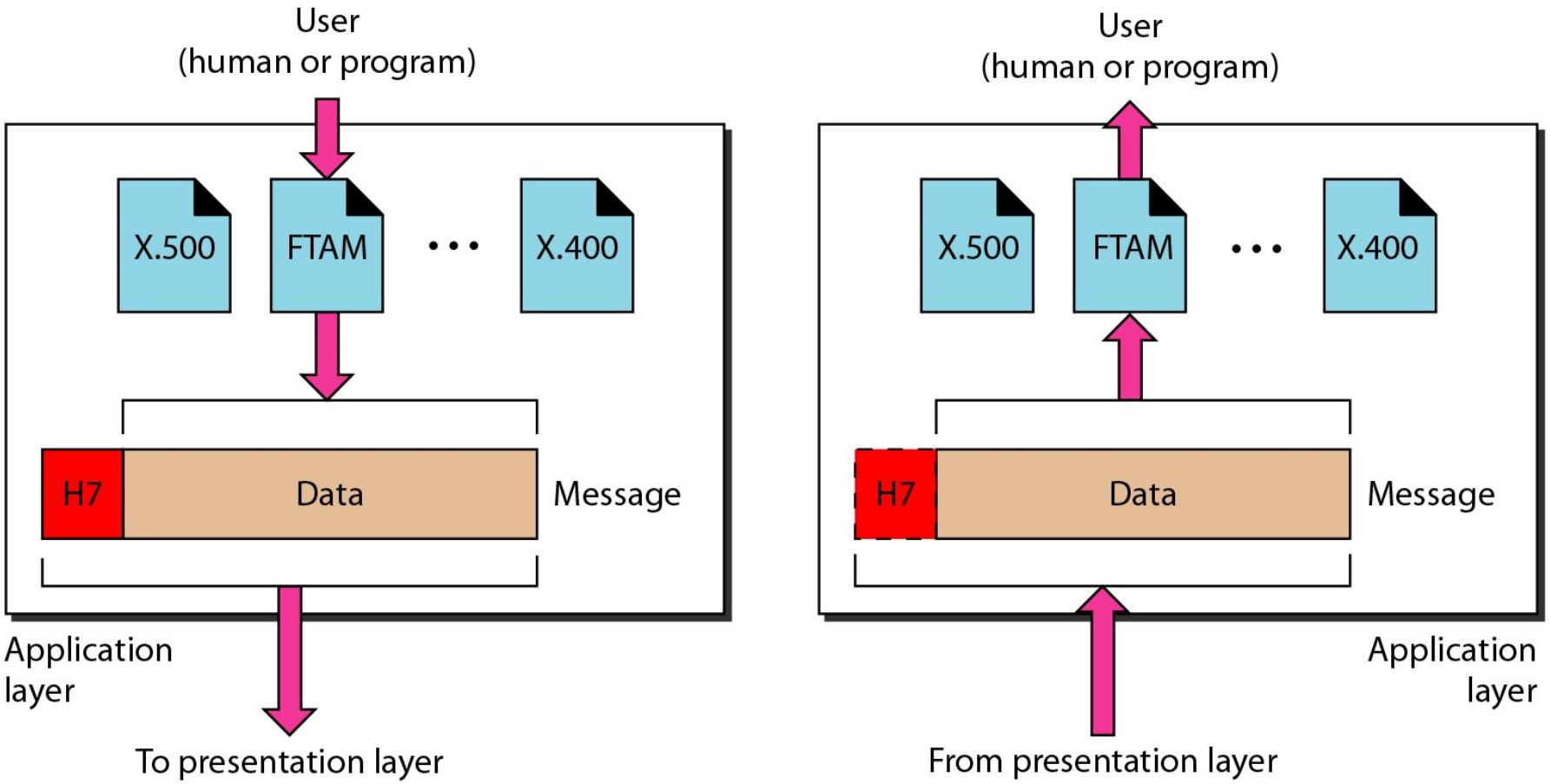




*Note*

**The presentation layer is responsible for translation, compression, and encryption.**

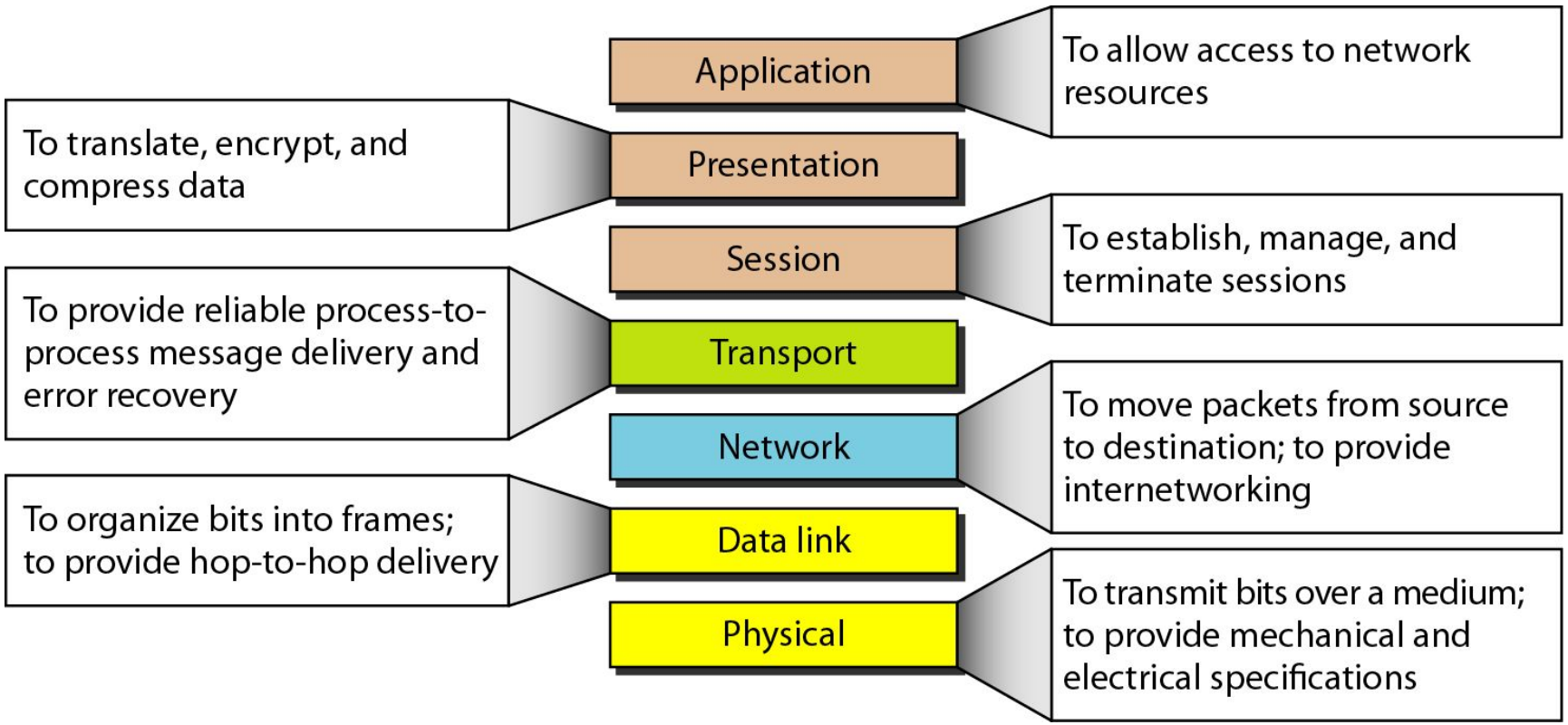
## Figure 2.14 *Application layer*



*Note*

**The application layer is responsible for providing services to the user.**

## Figure 2.15 *Summary of layers*



# Lack of OSI Model's Success

- TCP/IP protocol would be fully replaced by the OSI model. This did not happen for several reasons, but we describe only three, which are agreed upon by all experts in the field.

- First, OSI was completed when TCP/IP was fully in place and a lot of time and money had been spent on the suite; changing it would cost a lot.

- Second, some layers in the OSI model were never fully defined.
- Example, although the services provided by the presentation and the session layers were listed in the document, actual protocols for these two layers were not fully defined, nor were they fully described, and the corresponding software was not fully developed.

- Third, when OSI was implemented by an organization in a different application, it did not show a high enough level of performance to entice the Internet authority to switch from the TCP/IP protocol suite to the OSI model.

2.65

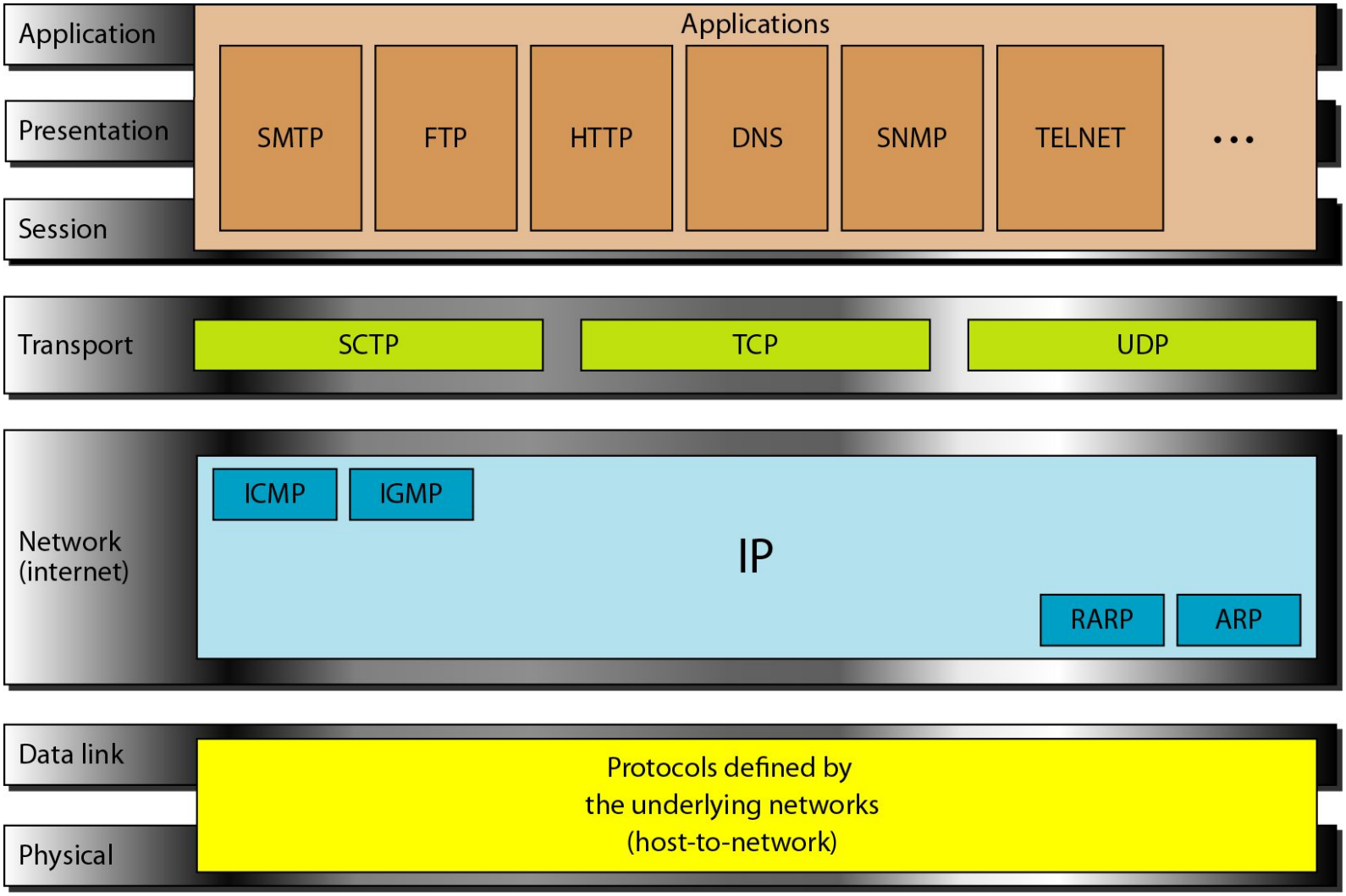
# 2-4 TCP/IP PROTOCOL SUITE

*The layers in the TCP/IP protocol suite do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers: host-to-network, internet, transport, and application. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: physical, data link, network, transport, and application.*

***Topics discussed in this section:***

**Physical and Data Link Layers**
**Network Layer**
**Transport Layer**
**Application Layer**

# Figure 2.16 *TCP/IP and OSI model*



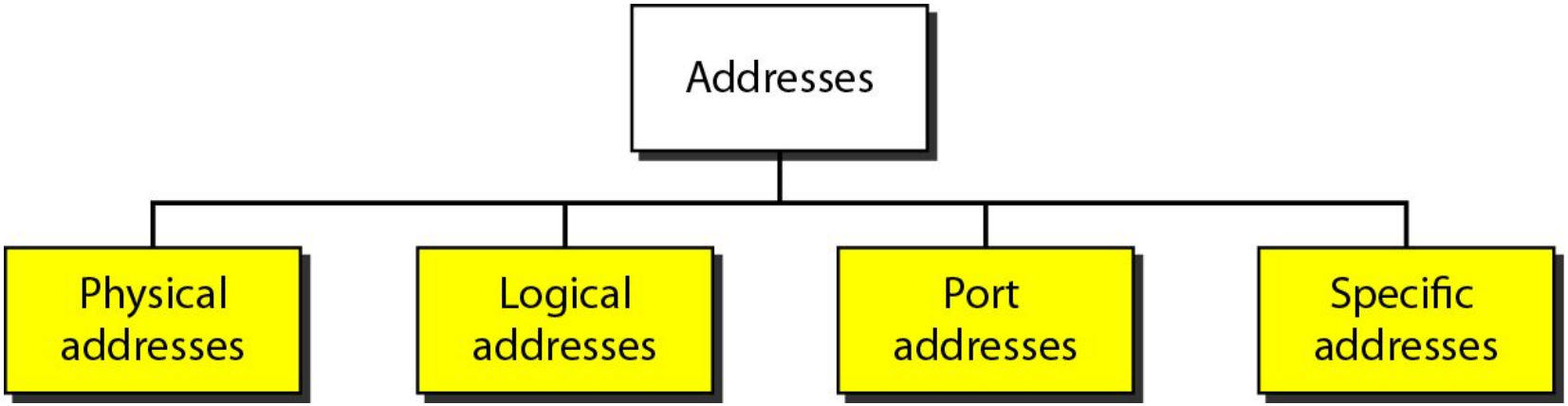
# 2-5 ADDRESSING

***Four levels of addresses are used in an internet employing the TCP/IP protocols: physical, logical, port, and specific.***

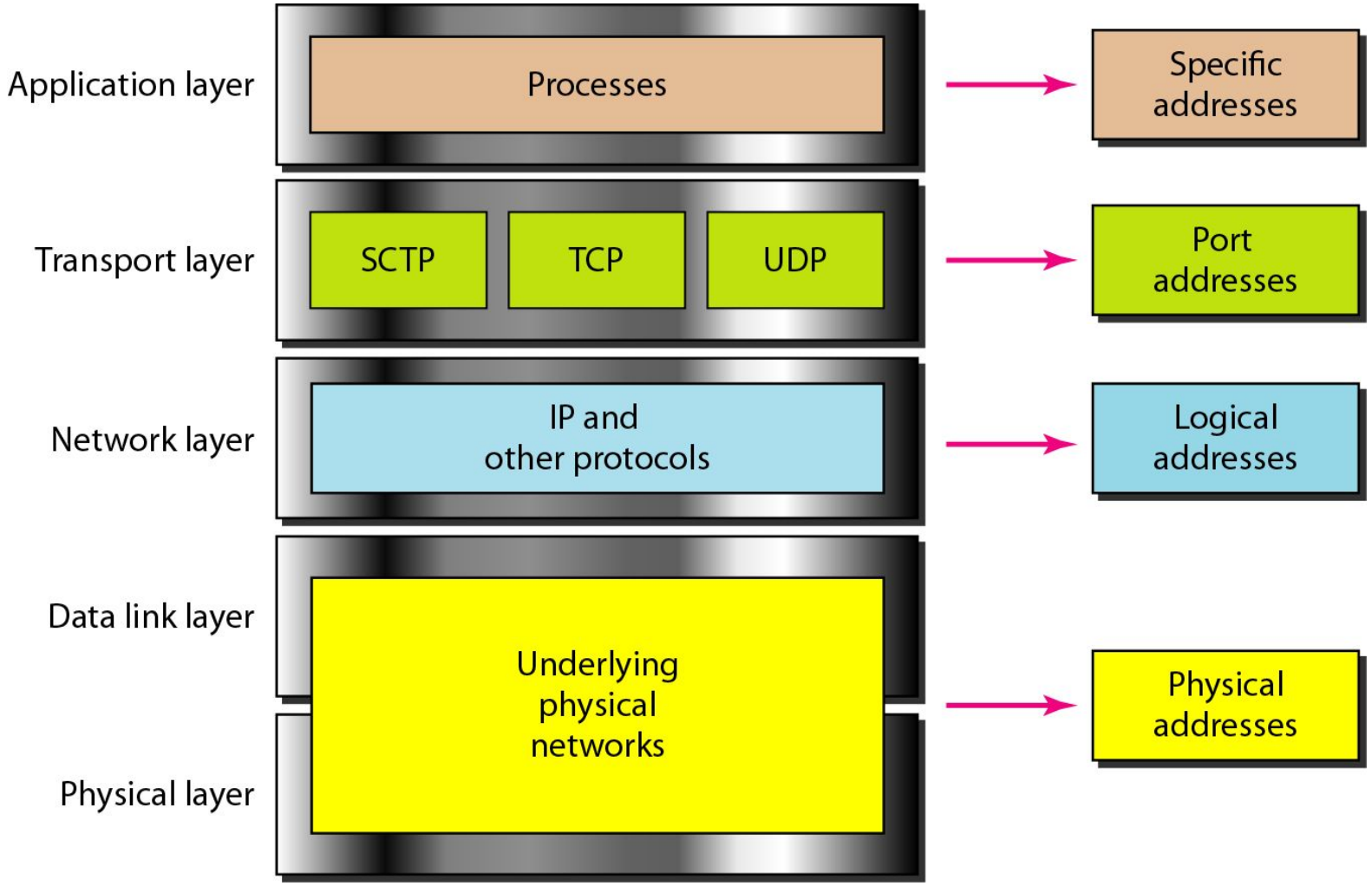
***Topics discussed in this section:***

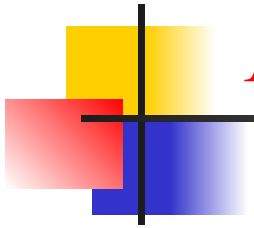
**Physical Addresses**
**Logical Addresses**
**Port Addresses**
**Specific Addresses**

## Figure 2.17 *Addresses in TCP/IP*



## Figure 2.18 *Relationship of layers and addresses in TCP/IP*

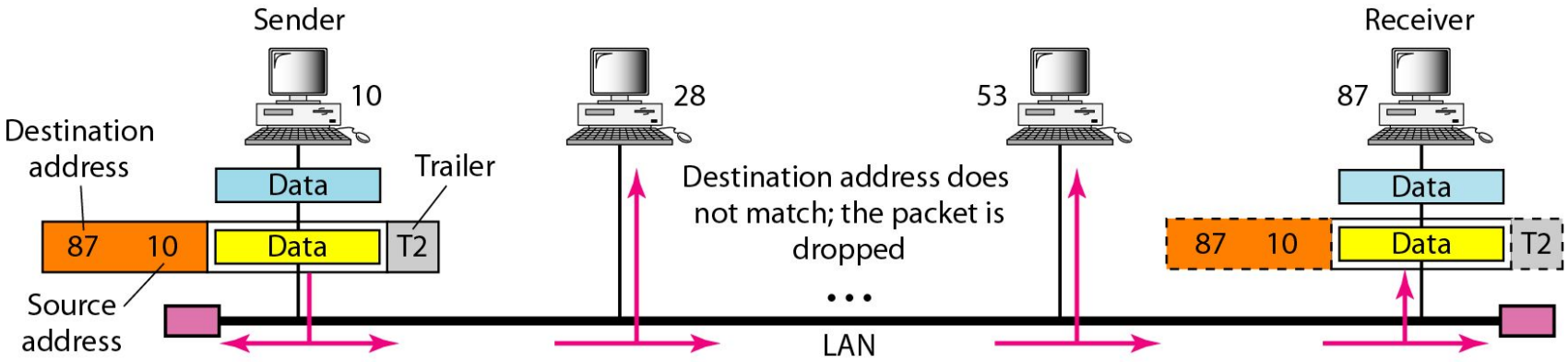




# *Example 2.1*

***In Figure 2.19 a node with physical address 10 sends a frame to a node with physical address 87. The two nodes are connected by a link (bus topology LAN). As the figure shows, the computer with physical address 10 is the sender, and the computer with physical address 87 is the receiver.***

# Figure 2.19 *Physical addresses*

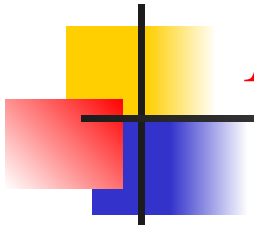


# *Example 2.2*

***Most local-area networks use a 48-bit (6-byte) physical address written as 12 hexadecimal digits; every byte (2 hexadecimal digits) is separated by a colon, as shown below:***

**07:01:02:01:2C:4B**

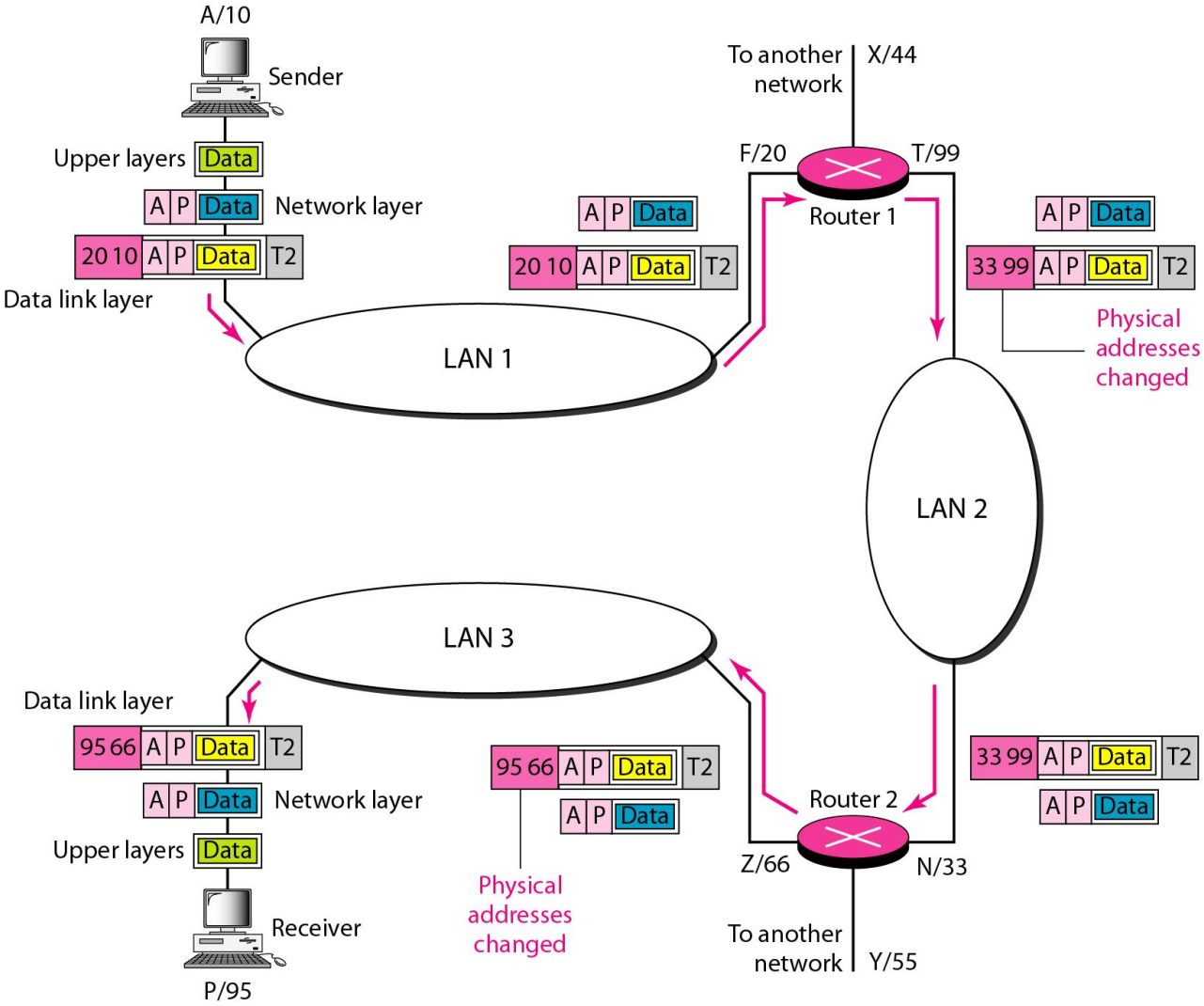
**A 6-byte (12 hexadecimal digits) physical address.**



# *Example 2.3*

***Figure 2.20 shows a part of an internet with two routers connecting three LANs. Each device (computer or router) has a pair of addresses (logical and physical) for each connection. In this case, each computer is connected to only one link and therefore has only one pair of addresses. Each router, however, is connected to three networks (only two are shown in the figure). So each router has three pairs of addresses, one for each connection.***

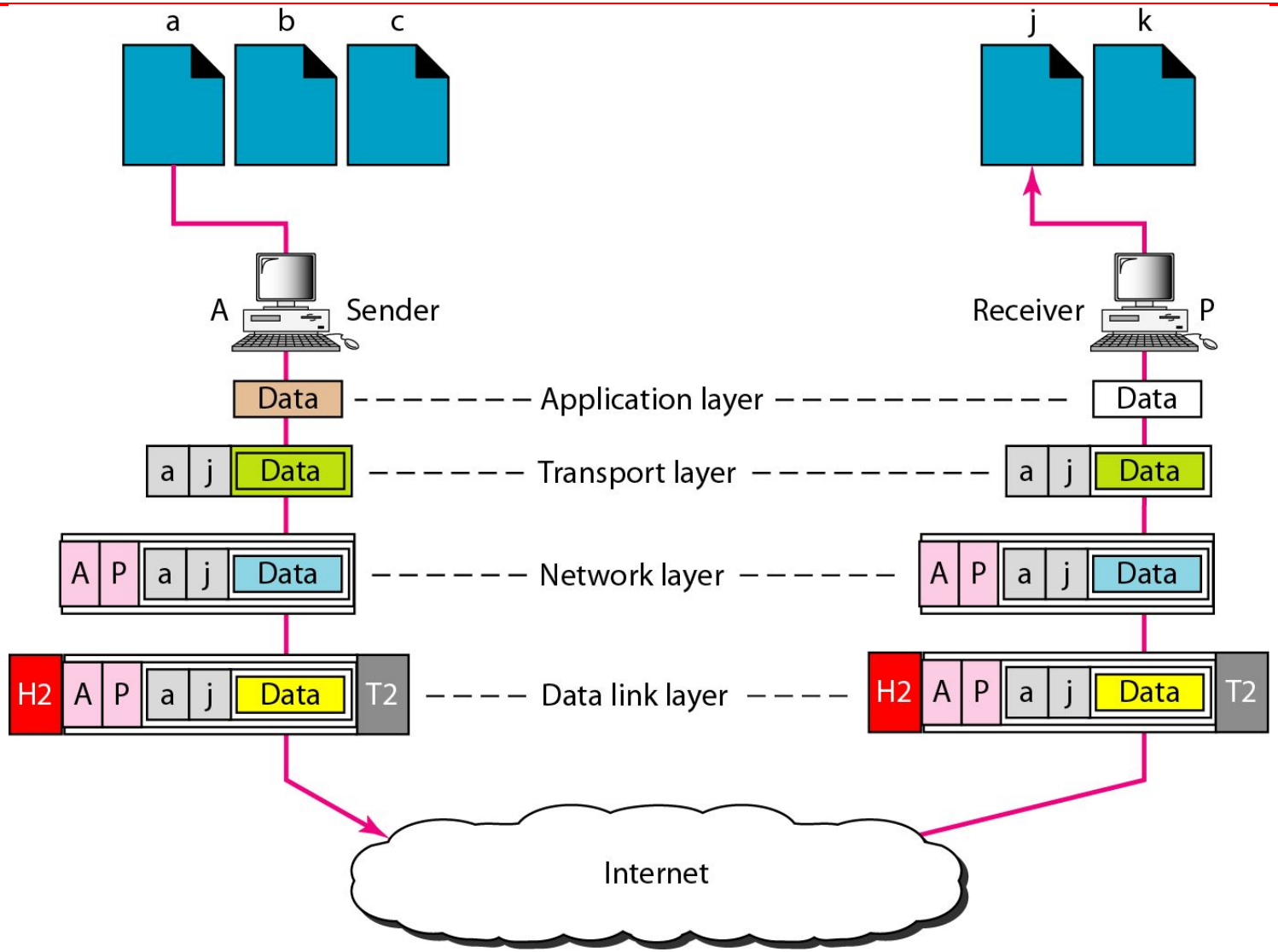
# Figure 2.20 *IP addresses*



# Example 2.4

*Figure 2.21 shows two computers communicating via the Internet. The sending computer is running three processes at this time with port addresses a, b, and c. The receiving computer is running two processes at this time with port addresses j and k. Process a in the sending computer needs to communicate with process j in the receiving computer. Note that although physical addresses change from hop to hop, logical and port addresses remain the same from the source to destination.*

# Figure 2.21 *Port addresses*



*Note*

**The physical addresses will change from hop to hop, but the logical addresses usually remain the same.**

# *Example 2.5*

***A port address is a 16-bit address represented by one decimal number as shown.***

**753**

**A 16-bit port address represented as one single number.**